

AI-Assisted Scientific Replication: A 30-Paper Evaluation

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Abstract

We report on a 30-paper evaluation of AI-agent-assisted replication of computational science papers drawn from the Argonne National Laboratory corpus on OSTI.gov, spanning 20+ domains (astrophysics, condensed matter, nuclear physics, computational chemistry, materials, power systems, computational biology, control theory, combinatorics, and more). Each replication was executed by an OpenAI GPT-family agent (Ollie / OpenClaw) with tool access and compute on CherryRd (local), uicgpu (shared 8×A100), Aurora, and Polaris. We scored each attempt on two independent 1–10 axes: *coverage* (what fraction of the paper’s contributions were reproduced) and *agreement* (how closely quantitative/qualitative results match). Across the cohort, mean coverage is 6.40/10 and mean agreement is 7.17/10. 8 papers scored ≥ 8 on both axes; 11 scored ≤ 5 on at least one axis. The strongest replications occurred where open-source code plus public data existed (combinatorics, graph algorithms, topological response), while the weakest involved proprietary pipelines (BV-BRC/SEEDtk genome binning), long HPC campaigns (CROC cosmic reionization), or paywalled/unpublished code (graph-RL distribution restoration). We identify recurring failure modes, map each underperforming paper to a concrete upgrade path, and distill 150 follow-on research questions (5 per paper) suitable for an AI-assisted research agent. The results support the thesis that, with appropriate tooling and compute, an AI agent can shoulder the mechanical burden of replicating a broad swath of published computational science and surface concrete extensions.

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1 Introduction

1.1 Motivation

Can an AI research agent, operating with modern LLM reasoning plus tool access and modest HPC, faithfully replicate published computational science? The question matters for three reasons: (i) reproducibility is the epistemic backbone of the field, yet wet- and dry-lab replication is expensive; (ii) agentic AI systems are now capable of non-trivial end-to-end pipelines, so their replication failure modes and strengths need empirical characterization; and (iii) a working replication pipeline opens the door to automated follow-on research — hypothesis generation, robustness probes, and scaling studies.

This report documents our first thirty attempts.

1.2 Scope and source

We drew papers from [OSTI.gov](https://www.osti.gov), filtered for Argonne-authored computational work with enough method detail to be reproducible. The cohort spans ~ 20 domains (Table 2): astrophysics/cosmology, nuclear physics, condensed matter, materials, computational chemistry, computational biology, quantum information, quantum optics, power systems, power electronics, combinatorics, graph algorithms, control theory, scientific ML, dark matter direct detection, combustion LES, and others. Publication years range from 2017 to 2026.

1.3 Methodology

Each replication was executed by an agentic pipeline consisting of: (a) **paper ingestion** (PDF $\rightarrow$ structured plan), (b) **environment setup** (uv/pip/conda, container images, data acquisition), (c) **implementation** (write or adapt code, usually in Python/Fortran/C++), (d) **execution** on the appropriate compute tier (CherryRd Mac Studio for small jobs; uicgpu 8 $\times$ A100 for mid-scale DL training; Aurora or Polaris for HPC ensembles), and (e) **scoring and reporting** against the published numbers. The model backbone used was primarily GPT-family (OpenAI) through an Argo proxy, with Anthropic Claude Opus used occasionally for writing and code review. Tool use was unrestricted within the workspace.

1.4 Scoring rubric

We scored each paper on two axes, following the schema in `scoring/SCHEMA.md` (reproduced in the Appendix):

- **Coverage** (1–10): fraction of paper’s contributions reproduced. 10 = every major element at or near paper’s scale; 8 = core method and main results, minor simplification; 6 = core method but significant scale/feature simplification; 4 = partial (e.g. comparator arm only); 2 = tangentially related demo; 1 = not replicated.
- **Agreement** (1–10): quantitative/qualitative fit to paper. 10 = within 5–10% everywhere; 8 = within 20% with correct trends; 6 = qualitative match, 20–50% metric gap; 4 = mechanism-level only; 2 = weak; 1 = contradicts.

We emphasize that the two axes are intentionally orthogonal: a small simplification of a paper’s architecture (low coverage) can still reproduce the main numerical finding exactly (high agreement), and a full-scale reimplementaion (high coverage) can diverge numerically due to force-field, stochastic-seed, or hyperparameter choices (low agreement).

2 Paper-by-Paper Evaluation

Papers are ordered by total score (coverage + agreement), descending. Ties are broken by agreement score.

2.1 1. Integer Sequences from Configurations in the Hausdorff Metric Geometry

OSTI: 1997354 **Authors:** Bobrowski, Elpers, Helmkamp, Ovsyannikov, Xique (2021) **Domain:** Pure Mathematics / Combinatorics

Coverage: 10/10 **Agreement:** 10/10 **Total:** 20/20

Coverage rationale. All 13 closed-form formulas from Theorems 8, 10, 11, 12 implemented and verified against brute-force bipartite-graph enumeration. All 21 OEIS sequences cross-referenced term-by-term. Exhaustive enumeration of 327,680 graphs up to K , and K , to confirm achievability of $[1,100]$. All five known non-achievable integers (19,37,41,59,67) confirmed. Fibonacci and Lucas achievability partially reconfirmed.

Agreement rationale. Every closed-form formula matches recursive enumeration exactly (integer arithmetic, no floating-point error). 21/21 OEIS sequences matched term-by-term. All 5 gaps confirmed. No discrepancies detected anywhere.

What reproduced:

- All 13 formula functions from Theorems 8, 10, 11, 12
- Brute-force bipartite edge-cover enumeration up to K , and K ,
- OEIS cross-referencing for all 21 sequences (A048291, A335608–A335613, A337416–A337418, A340173–A340175, A340199–A340201, A340897–A340899, A342580, A342796)
- Symmetry property $E(m,n)=E(n,m)$
- Five known non-achievable integer gaps (19,37,41,59,67) in $[1,100]$
- Fibonacci/Lucas achievability subsequences (8/10 and 5/6)

What missing:

- Extension beyond $m,n \leq 6$ to validate higher-order formulas at scale
- Complete resolution of the 35/100 unresolved integers in $[1,100]$
- Proof-level verification (we only verify numerically, not symbolically)
- Larger OEIS extensions to bound the next few non-achievable integers > 100
- Asymptotic density analysis of achievable integers
- Algorithmic characterization of non-achievability beyond brute force

Follow-on research questions:

- Q1. Extend the exhaustive graph enumeration to K , and K , to resolve the 35 open integers in $[1,100]$ — which remain non-achievable
- Q2. Is there a polynomial-time certificate for non-achievability, or does deciding achievability require exponential search Empirically test with SAT encoding on integers up to 200.
- Q3. Compute the asymptotic density of achievable integers in $[1,N]$ as $N \rightarrow \infty$ — does it tend to 1,

to a constant < 1 , or to 0

- Q4. Generalize the bijection to tripartite graph edge covers — what new integer sequences arise, and do they correspond to existing OEIS entries
- Q5. Which formula family (from Theorems 10–12) is tightest — i.e., attains the minimum $m+n$ realization for a given integer. Produce an atlas mapping each achievable integer to its cheapest bipartite realization.

2.2 2. Mathematical Foundations of the GraphBLAS

OSTI: 1379592 **Authors:** Kepner et al. (2016) **Domain:** Computer Science / Graph Algorithms

Coverage: 9/10 **Agreement:** 10/10 **Total:** 19/20

Coverage rationale. Complete from-scratch implementation of GraphBLAS algebraic foundations: Monoid, Semiring, SparseMatrix, SparseVector, plus 11 core operations (mxm, mxv, vxm, eWiseAdd, eWiseMult, extract, assign, apply, reduce, transpose, kronecker) generic over any user semiring with masking and accumulation. All 7 worked figure examples reproduced exactly. 6 classical graph algorithms (BFS, SSSP, triangles, betweenness, PageRank, CC) expressed purely via primitives. Only gap: no large-scale performance benchmarks vs SuiteSparse:GraphBLAS C reference.

Agreement rationale. Every worked figure matches exactly (12-edge adjacency, incidence products, BFS hop frontiers, etc.). All 4 quantitative algorithms (BFS, SSSP, triangles, PageRank) cross-validated against NetworkX on the paper’s 7-vertex graph and extra test graphs — all agree. 67/67 tests pass in $< 1s$. Since the paper is definitional, ‘agreement’ = internal consistency + algorithm correctness, both perfect.

What reproduced:

- Monoid/Semiring algebra with 7 built-in semirings (arithmetic, tropical, boolean, max-plus, min-times, max-min, plus-min)
- SparseMatrix/SparseVector with structural-zero semantics
- 11 core operations generic over any semiring
- Masking (structural, complement) and accumulation
- All 7 paper figure examples (Figs 1,2,4,5,6,7,8)
- 6 graph algorithms: BFS, SSSP, triangle count, betweenness, PageRank, connected components
- Cross-validation vs NetworkX on BFS, SSSP, triangles, PageRank

What missing:

- Large-scale performance benchmarks (paper is mostly definitional, so minor)
- Comparison vs SuiteSparse:GraphBLAS C reference implementation
- GPU-accelerated semiring operations
- Distributed / parallel semantics (HPC scaling)
- Streaming / incremental update operations
- Higher-order algorithms (max-flow, graph matching) beyond the core 6

Follow-on research questions:

- Q1. Benchmark our Python GraphBLAS vs SuiteSparse:GraphBLAS on SNAP graphs (LiveJournal, Orkut, Friendster) for BFS, SSSP, triangle count — what is the Python slowdown factor, and which primitives dominate
- Q2. Express three recent graph algorithms (GNN message-passing, graph spectral clustering, personalized PageRank with teleport) as GraphBLAS primitives — is the semiring abstraction sufficient, or what extensions are needed
- Q3. Design a semiring for uncertainty propagation (value + variance pairs) and express variance-aware BFS and SSSP — does the algebraic structure (associativity, distributivity) still hold, and does this yield provably-correct uncertainty quantification
- Q4. Compare GraphBLAS-expressed triangle counting against vertex-centric (Pregel) and matrix-free (SuiteSparse LAGraph) implementations on a 10-edge graph — at what scale does the matrix formulation beat vertex-centric, and where does each paradigm break
- Q5. Formally verify (e.g., in Coq or Lean) that our 6 graph algorithms are correct for any abstract semiring satisfying the stated axioms — this would provide the first machine-checked proof of GraphBLAS algorithm correctness.

2.3 3. Quantitative Relationship between Polarization Differences and the Zone-Averaged Shift Photocurrent

OSTI: 1523841 **Authors:** Fregoso, Morimoto, Moore (2017) **Domain:** Condensed Matter / Topological Response / Shift Current Photovoltaics
Coverage: 7/10 **Agreement:** 10/10 **Total:** 17/20

Coverage rationale. Reproduced the paper’s central analytical identity $e^*R_{\text{bar_cv}} = a^*(P_{\text{c}} - P_{\text{v}}) + W_{\text{cv}}e^*a$ (Eq. 9) on the Rice-Mele model (paper’s Appendix D testbed). Verified d-vector shift-vector formula (Eq. C5), analytic limit formulae (Eqs. D8-D9), Berry-connection closed forms (Eq. D4), and the winding-number discontinuity at the gap-closing point $\delta=0$. NOT replicated: the general multi-band formula (Section III/Eq. 17), material-specific DFT+Wannier calculations (GeS, BaTiO₃, etc.), nor the frequency-integrated shift-conductivity connection.

Agreement rationale. Identity verified to machine precision: residual $|e^*R_{\text{bar_cv}} - a^*(P_{\text{c}} - P_{\text{v}})| \sim 3.3e-16$ pointwise on a 100-point δ sweep. Analytic limit formulas reproduced to $\sim 1e-16$. Winding number $|W_{\text{cv}}|=1$ correctly identified with $\Delta W=1$ jump across $\delta=0$. This is essentially a perfect analytical replication on the 1D two-band model the paper uses as its primary illustration.

What reproduced:

- Paper Eq. 9: $e^*R_{\text{bar_cv}} = a(P_{\text{c}} - P_{\text{v}}) + W_{\text{cv}}e^*a$
- Paper Eq. C5: d-vector shift-vector closed form
- Paper Eqs. D8-D9: analytic $k \rightarrow 0$ and $k \rightarrow \pi/a$ limits
- Paper Eq. D4: Berry connections $A_{\text{nn}}(k)$
- Winding-number integer discontinuity at gap closing
- Fig. 1(b) qualitative structure: $P_{\text{c}} - P_{\text{v}}$ vs d-vec vs full numerical R_{bar}

- Gauge-unique closed-form Berry-connection integration

What missing:

- General N-band formula (Section III / Eq. 17)
- Material-specific first-principles + Wannier calculations
- Frequency-integrated shift-conductivity connection
- Finite-frequency shift current spectra $\hat{\sigma}^{\text{shift}}(\omega)$
- Paper’s Figs. 2-4 (material examples, higher-D extensions)
- Comparison against experimental shift-current measurements

Follow-on research questions:

- Q1. Extend the numerical verification to a 3-band or 4-band tight-binding model (e.g. SSH+onsite, or a minimal GeS-like model) and check whether the identity generalizes via the Wilson-loop formulation of Eq. 17.
- Q2. Compute the shift-current conductivity $\hat{\sigma}^{\text{shift}}(\omega)$ in the Rice-Mele model and verify the frequency integral equals $(\pi^2 e^3 / \hbar^2) (P_c - P_v)$ modulo the winding contribution.
- Q3. Apply the identity as a DFT+Wannier post-processing tool for a material like GeS or monolayer WS₂: compute $P_c - P_v$ from Wannier centers and compare to the direct frequency-integrated $\hat{\sigma}^{\text{shift}}$.
- Q4. Study how the winding number W_{cv} changes under SHG-relevant perturbations (strain, electric field) and predict a shift-current discontinuity as a material tuning knob.
- Q5. Generalize to interacting systems: does the identity survive a Hartree-Fock or GW correction when $P_c - P_v$ is evaluated with quasiparticle wavefunctions Test on a Hubbard-Rice-Mele numerical example.

2.4 4. Exactly-Solved Model of Light-Scattering Errors in Quantum Simulations with Metastable Trapped-Ion Qubits

OSTI: 2441075 **Authors:** Foss-Feig, Vikas, et al. (2024) **Domain:** Quantum Information / AMO

Coverage: 8/10 **Agreement:** 9/10 **Total:** 17/20

Coverage rationale. Complete implementation of analytic solution (Eqs. 6-10) with all four terms I, R, L, B — including the paper’s novel leakage (L) and combined (B) contributions. Five-jump-operator Lindblad master equation implemented numerically with correct factor-of-2 jump convention. Ca branching ratios (94.5% leakage / 3.9% Raman / 1.6% elastic). Figures 1 (schematic), 2 (GHZ fidelity), 3 (correlations), 4 (squeezing) reproduced qualitatively. Missing: full Penning-trap 2D crystal mode computation (we used 1D chain approximation for J_{ij}).

Agreement rationale. Analytic-numerical agreement at machine precision (errors 10^{-10} across all tested cases). Trace preservation to 5.5×10 , positivity to 10 — essentially exact. All qualitative figure features correct: fidelity with N, Postselection helps, m-body correlations decay faster than $\exp(-mt)$, $N^{-2/3}$ scaling with scattering floor. Quantitative figure values differ because paper’s Penning-trap J_{ij} differs from our 1D-chain J_{ij} .

What reproduced:

- Full analytic solution Eqs. 6-10 with all four I, R, L, B components (including the paper’s novel L and B terms)
- 5-jump-operator Lindblad numerical master equation with correct factor-of-2 jump convention
- Ca branching ratios (94.5% leakage / 3.9% Raman / 1.6% elastic)
- Selection rule: $|1\rangle = |m_J = -5/2\rangle$ dark under π polarization
- Machine-precision analytic vs numerical agreement (10^{-10} – 10^{-3})
- Figure 2 GHZ-fidelity qualitative trends (N dependence, postselection gain, B-field effect)
- Figure 3 correlation decay (faster than $\exp(-mt)$) and perpendicular-correlation growth
- Figure 4 spin-squeezing $\propto N^{-2/3}$ with scattering floor

What missing:

- Penning-trap 2D triangular-lattice ion-crystal mode structure
- Full 3D normal-mode J_{ij} computation (we used 1D chain)
- Laser power P and beam waist w (not specified in paper)
- Exact trap frequencies ω_z, ω_r (referenced to Ref [20])
- GHZ fidelity fourth-order δJ_{ij} correction terms
- Quantitative N-thresholds from Fig 2 match
- Experimental validation against real Ba or Ca data

Follow-on research questions:

- Q1. Implement the full 2D Penning-trap equilibrium-crystal mode solver and recompute J_{ij} — do the Fig 2 fidelity crossovers match the paper’s reported $N_{\text{threshold}}$ at $B=0.9\text{T}$ and $B=4.5\text{T}$ within 10%
- Q2. Extend the exact solution from π to $\sigma\pm$ polarization where $|1\rangle$ is no longer dark — derive new analytic expressions and quantify the fidelity penalty of losing the dark-state protection.
- Q3. Test the exact solution against a full Lindblad simulation at $N=20$ (near the boundary of tractable numerics) — does the product-form correlation factorization still agree to machine precision, or do many-body corrections emerge
- Q4. For ^3Ba (different branching ratios, $\sim 70\%$ leakage), do the novel L and B terms still dominate the error budget, and does postselection recover the same 5.5% residual-error advantage
- Q5. Combine the exact scattering model with realistic single-qubit gate errors (miscalibration, Stark shifts) to produce a full end-to-end error budget for a 50-qubit GHZ preparation circuit — at what N does gate error overtake scattering as the dominant error source

2.5 5. Clustering huge protein sequence sets in linear time

OSTI: 1624105 **Authors:** Steinegger & Soeding (2018) **Domain:** Computational biology / algorithms

Coverage: 8/10 **Agreement:** 9/10 **Total:** 17/20

Coverage rationale. From-scratch Python reimplement of Linclust’s three-stage algorithm

(bottom-m k-mer sketch, bucketed center assignment, ungapped identity verification) with a matched $O(N \log N)$ complexity test. Production MMseqs2 binary was not rerun on UniRef-scale datasets (memory footprint, cluster-count and compression ratios on $>1e8$ sequences) due to the 2-hour compute budget; we benchmarked up to $N=64k$ synthetic proteins.

Agreement rationale. Algorithmic claim reproduces directly: measured log-log slope 1.31 (linclust) vs 2.08 (naive $O(N^2)$) — the paper’s linear-time signature. $\sim 240x$ speedup at $N=4000$ and $F1 \geq 0.987$ vs ground truth both match the paper’s qualitative performance claims (precision + linear scaling). Absolute wall-time numbers will differ from the paper’s C++ MMseqs2 code, but the scaling exponents agree to within expected constant factors.

What reproduced:

- Bottom-m k-mer sketching as the locality-sensitive primitive
- Bucketed center-assignment strategy
- Ungapped identity verification stage
- Near-linear scaling empirically confirmed on synthetic proteins up to $N=64k$
- Cluster-quality $F1 \geq 0.987$ vs ground truth
- Scaling and quality figures (log-log and linear axes)

What missing:

- Production MMseqs2 C++ binary run on UniRef50/UniRef100 ($1e8$ - $1e9$ sequences)
- Paper’s reported wall-time and memory footprint on 1.6B sequences
- Cluster count / compression-ratio comparison with CD-HIT / UCLUST / kClust
- Pre-filter vs ungapped-alignment vs Smith-Waterman sensitivity curves
- Cascaded clustering (linclust $\rightarrow$ mmseqs2 iterative) comparison

Follow-on research questions:

- Q1. How does Linclust’s clustering quality ($F1$ vs curated reference families) degrade as intra-family identity is pushed from 50% down to 30% on a controlled synthetic benchmark
- Q2. Can the bottom-m sketch be replaced with a learned MinHash via a small Transformer embedding, and does that improve remote-homolog recall at matched runtime
- Q3. What is the GPU speedup achievable by porting the bucketed center-assignment stage to CUDA (hash-table on device memory), and does the memory bandwidth bound dominate
- Q4. How sensitive is the linear-time scaling exponent to the k-mer length k and the sketch width m , and is there a regime where the method silently becomes $O(N \log N)$
- Q5. When inputs are metagenome-scale ($>5e9$ sequences with heavy sequence redundancy), does Linclust’s single-pass center selection still produce stable cluster boundaries under random input shuffling

2.6 6. Common Mode Voltage Reduction of Single-Phase Quasi-Z-Source Inverter-Based Photovoltaic System

OSTI: 1606674 **Authors:** Sajadian, Ahmadi, Kouzani (2019) **Domain:** Power electronics
Coverage: 8/10 **Agreement:** 9/10 **Total:** 17/20

Coverage rationale. Independent NumPy SPWM generator plus PySpice/ngspice transient simulation of the common-mode equivalent circuit (Eqs 5-6 of the paper) with the same modulation indices, switching frequency, shoot-through duty, and CM-path R/L/C values. Proposed ZS1->ZS2 scheme and traditional scheme both implemented. Not replicated: hardware prototype measurements, MPPT loop, full dq grid-synchronisation block, EMI filter design, component-tolerance Monte Carlo.

Agreement rationale. CMV halving reproduces exactly: $|U_{CM}|_{peak} 400\text{ V} \rightarrow 200\text{ V}$ (50% reduction), matching the paper's central claim that proposed scheme clamps CMV to $\{0, V_{PN}/2\}$ while traditional scheme visits $\{0, V_{PN}/2, V_{PN}\}$. Leakage current peak 40.7% reduction (730 mA $\rightarrow$ 433 mA) and RMS 36.7% reduction align with the paper's reported envelope (paper reports leakage in the same ballpark; exact match not expected without full CM-path parasitics). Switching-harmonic line at 10 kHz drops by ~ 40 dB in our spectrum, consistent with paper Fig. 8(c).

What reproduced:

- Unipolar SPWM at $M=0.78$, $f_s=10\text{ kHz}$
- Shoot-through duty $D=0.125$ qZSI modulation
- Common-mode equivalent circuit ($L_f=3\text{ mH}$, $C_g=2.2\text{ nF}$, $R_g=100\text{ Ohm}$, $R_f=2\text{ Ohm}$)
- Traditional vs proposed modulation leg-voltage traces
- 50% CMV peak reduction (exact)
- ~ 40 dB CMV switching-harmonic attenuation at 10 kHz
- Leakage-current waveform comparison (Fig. 6c vs 8c)

What missing:

- Experimental hardware prototype
- MPPT and grid-synchronisation control loops
- Full EMI filter stage
- Efficiency and THD numbers
- Component-tolerance sensitivity study
- Thermal / conduction-loss analysis

Follow-on research questions:

- Q1. How does the 50% CMV reduction degrade as the CM parasitic capacitance C_g varies over 0.5-10 nF (representative of different PV-array sizes and mounting conditions)
- Q2. Does the ZS1->ZS2 modulation preserve its CMV advantage under dead-time insertion of 0.5-2 μs , or does dead-time reintroduce intermediate CM levels
- Q3. Can a third modulation state (ST->ZS3) further reduce CMV peak below $V_{PN}/4$ without

penalising DC-link utilisation, and what is the achievable theoretical floor

- Q4. How do the CMV and leakage-current reductions translate to measured conducted EMI spectra (150 kHz-30 MHz) on a prototype, and does a smaller EMI filter become viable
- Q5. When the grid voltage is distorted ($>5\%$ THD, notched by nonlinear loads), does the proposed scheme maintain the clean two-level CMV, or does the control interaction reintroduce transients

2.7 7. Markov State Models from Short Non-Equilibrium Trajectories via Observable Operator Models

OSTI: 1565592-MSM-Hempel **Authors:** Nüske, Wu, Wehmeyer, Clementi, Noé (2017) **Domain:** Computational Chemistry / MSM theory

Coverage: 9/10 **Agreement:** 8/10 **Total:** 17/20

Coverage rationale. All three numerical experiments replicated: 1D double-well, 2D potential, and alanine dipeptide (ADP) with $11,388 \times 20$ ps short trajectories plus 1 μ s long reference on 8×A100 OpenMM. OOM correction implemented and tested end-to-end. Only gap: limited exploration of τ -sweep and basis-function variations reported in supplementary.

Agreement rationale. Phase 1 matches paper to 0.2% (t 3699 vs 3708). Phase 2 within 2%. Phase 3 ADP: direct MSM underestimates t (299 ps), OOM corrects to 2146 ps — $7.2\times$ improvement, matching 2020 ps long-reference within 6%. Absolute t differs from paper’s ~ 1400 ps by $\sim 44\%$, attributed to force-field differences (AMBER ff14SB vs paper’s). Replication also discovered a practical subtlety (clustering must cover both basins) not explicit in paper.

What reproduced:

- 1D double-well exact timescale via transfer operator spectrum
- Direct MSM bias demonstration and OOM correction (Phase 1,2,3)
- ADP simulation at $11,388 \times 20$ ps scale on 8×A100 with AMBER ff14SB + TIP3P
- Long-reference 10×100 ns ground truth for ADP timescales
- OOM-based timescale recovery at $\tau=5$ ps (299→2146 ps correction)
- k-means clustering + MSM construction pipeline

What missing:

- Exact force field reproduction (we used ff14SB, paper’s choice unclear)
- Full τ -sweep showing OOM robustness across lag times
- Supplementary basis-function variants (TICA vs k-means)
- Paper’s exact $t \approx 1400$ ps — our 2020 ps reference differs (force field origin)
- Explicit stationary-distribution reweighting benchmarks
- Finite-sample uncertainty quantification across many trajectory subsets

Follow-on research questions:

- Q1. Is the 44% t gap (2020 vs 1400 ps) attributable to force field, water model, or integrator — test by running identical short-trajectory protocol with CHARMM36m and GROMOS54a7 and comparing long-reference t .

- Q2. How does OOM correction scale to ≥ 3 metastable states Apply to chignolin or villin headpiece where ≥ 3 slow modes exist and compare OOM vs direct MSM recovery of t , t , t .
- Q3. Does the 'both-basin-coverage' clustering requirement we discovered translate into a principled seeding algorithm (e.g., farthest-point after long-ref preseeding) that recovers OOM accuracy without requiring a long reference
- Q4. Benchmark OOM vs Koopman-based VAMPnets on identical short-trajectory ADP data — which method recovers t with fewer trajectories, and how does the crossover depend on trajectory length
- Q5. Quantify the breaking point: at what short-trajectory length $T_{\min}$ does OOM no longer correct the MSM bias (error $> 20\%$ of long-ref t) Sweep $T \in \{1,5,10,20,50\}$ ps at fixed total data.

2.8 8. Approximating Photo-z PDFs for Large Surveys

OSTI: 1461824 **Authors:** Malz & Marshall (2018) **Domain:** Astrophysics / Statistical Compression

Coverage: 8/10 **Agreement:** 8/10 **Total:** 16/20

Coverage rationale. All three formats (histogram, samples, quantiles) implemented with paper-matching reconstruction (piecewise-constant, Gaussian KDE, PCHIP spline-CDF). Protocol matched exactly: $N_f \in \{3,10,30,100\}$, 10 random subsamples of 100 galaxies each, KLD + moment errors ($m=1,2,3$). Bright and faint synthetic catalogs (100K galaxies each) with 3/5-component GMMs. Main gap: mock catalogs are synthetic GMMs, not BPZ-processed Buzzard/Millennium simulations (not publicly available).

Agreement rationale. All qualitative claims confirmed: (1) quantiles best for broad/multimodal PDFs (KLD=0.001 vs 0.052 for histogram at $N_f=100$, faint); (2) histograms best for narrow/unimodal (bright); (3) samples least efficient. Moment errors track KLD. Monotonic improvement with N_f . Absolute KLD values same order of magnitude as paper; exact numerical comparison limited by synthetic vs. BPZ catalog differences.

What reproduced:

- Three format implementations with matching reconstruction methods
- N_f grid $\{3, 10, 30, 100\}$
- 10 random subsamples $\times$ 100 galaxies protocol
- KL divergence on fine grid with floor $\epsilon=10^3$
- Moment errors Δ , Δ , Δ (mean, variance, skewness)
- Stacked $n(z)$ comparisons at all N_f
- Qualitative ordering: quantiles histograms $<$ samples on faint; histograms $<$ quantiles $<$ samples on bright

What missing:

- Actual BPZ-processed mock catalogs (Graham et al. 2018 / Buzzard / Millennium)
- Realistic photometric-noise distribution of PDF shapes
- Paper's custom quantile-spline with linear tail extrapolation (we used PCHIP)

- Specified KDE bandwidth (we used scipy Scott’s rule)
- Full qp-package integration
- Storage-vs-accuracy Pareto frontier in bytes (not just N_f)
- Downstream cosmology-inference impact test

Follow-on research questions:

- Q1. Generate BPZ-like mocks from a real photometric catalog (e.g., COSMOS2020 or HSC-SSP) and re-run the KLD benchmark — does the quantile dominance hold when the PDF shape distribution comes from real data
- Q2. At fixed byte budget (not N_f), which format wins Compute exact byte cost including meta-parameters and float precision (float16 vs float32) — does quantiles’ advantage shrink when samples use int16 indices
- Q3. Train a neural PDF compressor (VAE or normalizing flow) on the same mock catalogs — does a learned 10-dim latent beat the best classical format at matched byte budget
- Q4. How does quantile reconstruction degrade under realistic tail truncation (z quantile levels clipped to $[0.01, 0.99]$) Sweep clipping thresholds and identify the regime where samples format surpasses quantiles.
- Q5. Propagate each compressed format through a 2-point cosmic-shear cosmological inference — does the quantile format’s lower KLD translate to tighter Ω_m, σ contours, and by how many percent

2.9 9. Motion Tomography via Occupation Kernels

OSTI: 1842593 **Authors:** Christensen et al. (2021) **Domain:** Control Theory / Kernel Methods

Coverage: 8/10 **Agreement:** 8/10 **Total:** 16/20

Coverage rationale. Full Algorithm 1 implementation (iterative flow-field reconstruction via occupation-kernel Gram-matrix inversion). Three synthetic flow fields (Gaussian mixture, linear/spiral, constant) tested. Parameter sensitivity sweeps over μ (kernel width), λ (regularization), N (number of trajectories). 26/26 tests pass covering kernel properties, Gram-matrix structure, and reconstruction convergence. Main gap: Experiment 2 (real-world Gliderpalooza dataset from Chang et al. [5]) not replicated — no data access. No comparison with alternative RKHS-based or deep-learning flow reconstruction methods.

Agreement rationale. Algorithm converges with 5-order-of-magnitude displacement reduction over 10 iterations ($0.265 \rightarrow 5 \times 10$). Mean relative error 0.0155 (paper: 0.0256) — ours slightly better, attributed to different random trajectory seeds. Convergence ordering Constant < Linear < Gaussian confirmed. U-shaped μ sensitivity with optimum ≈ 0.5 confirmed. N sweep shows monotonic improvement with more trajectories. λ sensitivity monotonically increasing (paper’s was U-shaped, likely because of different Gram-matrix conditioning).

What reproduced:

- Algorithm 1 iterative flow reconstruction via occupation-kernel RKHS
- Gaussian RBF and exponential-dot-product kernels

- Occupation-kernel evaluation via Simpson’s rule
- Gram-matrix computation via 2D Simpson’s rule
- All three synthetic flow fields from the paper
- Parameter sweeps over kernel width μ , regularization λ , trajectory count N
- Convergence ordering and rates for the three fields
- Error metric (mean relative error) within paper’s range

What missing:

- Experiment 2: Real-world Gliderpalooza dataset from Chang et al. [5]
- Comparison with alternative flow-reconstruction methods (sparse regression, GP, NN)
- Theoretical convergence-rate verification against Theorems 2.3, 3.1
- Robustness to observation noise in endpoint measurements
- Non-Gaussian kernel families beyond RBF and exponential-dot-product
- Scaling analysis: wall-clock vs N (our $O(N^2 \cdot n^2)$ cost model)
- Time-varying flow fields $F(x,t)$

Follow-on research questions:

- Q1. Download a public oceanographic glider dataset (e.g., IOOS Glider DAC) and apply Algorithm 1 to real endpoint displacements — does kernel width $\mu=0.5$ remain optimal on physical-scale flows, and what is the reconstruction error vs HYCOM model ground truth
- Q2. Add Gaussian observation noise σ to endpoint displacements and characterize algorithm breakdown — at what σ /displacement ratio does the reconstruction error exceed 20%, and does Tikhonov regularization extend the tolerable noise range
- Q3. Replace the occupation-kernel RKHS with a divergence-free neural-network parameterization — does the learned approach beat the kernel method on the Gaussian-mixture flow at $N=25$, and how does it scale to 4D (spatial+time) flows
- Q4. Extend Algorithm 1 to time-varying flows $F(x,t)$ via space-time kernels $K((x,t),(y,s))$ — does it recover a known oscillating-vortex test case, and what is the effective temporal resolution as a function of trajectory density
- Q5. For the Gaussian-mixture flow, characterize identifiability: with fixed trajectory endpoints, what subspace of F is unrecoverable, and does this null space correspond to a closed-form RKHS orthogonal complement

2.10 10. Chiral Spin Order in Kondo-Heisenberg Systems

OSTI: 1412756 **Authors:** Unknown (OSTI 1412756) (2017) **Domain:** Condensed Matter Theory / Strongly Correlated Electrons

Coverage: 7/10 **Agreement:** 8/10 **Total:** 15/20

Coverage rationale. Reproduced the paper’s analytical mean-field backbone: zero-T energy functional $E_0(\alpha)$, critical Heisenberg coupling J_c from $C(J_c)=1$ condition, three-regime phase progression (disordered / CSL / AFM), finite-T Ising reduction, and scalar-chirality CSL

dome. NOT replicated: rigorous 2D Ising universality ($\beta=1/8$) via field theory or Monte Carlo; lattice-specific triangular/kagome applications; material-specific calculation for $\text{Sr}_2\text{VO}_3\text{FeAs}$.

Agreement rationale. J_c matches analytical Eq. (Jc) to machine precision. Canting angle $\alpha^*(J_H)$ shows all three regimes. Quadratic $T_c \sim (J_H - J_c)^2/J_K$ captured near threshold. Phase boundary qualitatively matches paper Fig. 3. Main discrepancy: our mean-field Ising reduction gives weakly first-order transition in some parameter ranges vs paper's claimed continuous 2D Ising transition ($\beta=1/8$) — acknowledged honestly as MF artifact.

What reproduced:

- Energy functional $E_0(\alpha)$ (paper Eq. E0)
- Critical coupling J_c via $C(J_c)=1$ (paper Eq. Jc)
- Canting angle $\sin(\alpha^*)$ vs J_H/J_c
- Finite-T Ising order parameter $m(T)$ and $T_c(J_H)$
- Quadratic onset near threshold: $T_c \sim (J_H - J_c)^2$
- Scalar chirality $O_c \sim s^3 \sin(\alpha) \cos^2(\alpha)$ CSL dome
- Z_2 (time-reversal x parity) symmetry breaking with $SU(2)$ intact

What missing:

- 2D Ising critical exponent $\beta=1/8$ (requires MC / field theory)
- Lattice-specific triangular and kagome extensions
- Material-specific $\text{Sr}_2\text{VO}_3\text{FeAs}$ numbers
- Berry curvature / anomalous Hall transport predictions
- Fluctuation corrections beyond mean-field saddle-point

Follow-on research questions:

- Q1. Run a classical Monte Carlo on the Ising-alpha effective model (on a 2D lattice with coarse-grained coherence volumes) to numerically extract β and test the claimed 2D Ising universality.
- Q2. Generalize the mean-field ansatz to triangular and kagome Kondo-Heisenberg lattices and compute how the CSL dome shifts — does kagome frustration widen the CSL window between J_c and J_{AFM}
- Q3. Incorporate itinerant-electron Berry curvature in the CSL phase and predict the anomalous Hall conductivity $\sigma_{xy}(T)$ below T_c ; compare to measured σ_{xy} in candidate chiral-spin materials.
- Q4. Relax the nested-FS assumption ($\tilde{J}_H(Q)=0$) and scan filling to map how the CSL dome deforms away from perfect nesting; identify the filling window where CSL survives.
- Q5. Couple the mean-field electronic structure to phonons and ask whether the CSL onset couples to a lattice distortion (magnetoelastic signature) observable in Bragg-peak splitting.

2.11 11. Molten Salt Reactor Neutronics and Fuel Cycle Modeling and Simulation with SCALE (MSRE Depletion)

OSTI: 2217719 **Authors:** Betzler, Powers, Worrall (2017) **Domain:** Nuclear Engineering / Reactor Physics

Coverage: 6/10 **Agreement:** 8/10 **Total:** 14/20

Coverage rationale. Reproduced MSRE core geometry (2D axial slice, 22.5% fuel fraction), fuel salt composition, graphite moderator, Hastelloy-N vessel, and online noble-gas/noble-metal removal rates. Used OpenMC Monte Carlo depletion with 25 steps x 15 days = 375 days. Did NOT replicate: full 3D deterministic SCALE/TRITON model, full ~900-day operational history with fuel additions, or the broader fuel-cycle sensitivity studies of the paper.

Agreement rationale. Initial $k_{\text{eff}} = 1.165 \pm 0.001$ (ours) vs ~1.16 (paper SCALE) — excellent agreement to ~0.5%. Leakage 18.1% vs paper 15-20% range. Reactivity decreases monotonically as expected ($\Delta k = 0.077$ over 375 d). U-235 depletion and Pu-239 buildup match expected magnitudes. No direct depletion-curve overlay in report, but key quantities align.

What reproduced:

- MSRE core geometry (hex lattice, graphite stringers, core can, reflector)
- LiF-BeF₂-ZrF₄-UF₄ fuel salt composition at 2.32 g/cm³, 922 K
- Initial $k_{\text{eff}} \sim 1.165$ matching paper
- Online processing rates: noble gas 4.067e-5/s, noble metal 8.667e-3/s
- OpenMC PredictorIntegrator depletion over 375 days
- Fuel fraction 22.5%, leakage ~18%

What missing:

- Full 3D SCALE/TRITON deterministic model with 1-group transport
- Full ~900-day operational history with fuel additions / U-235 makeup
- Broader fuel-cycle parameter sweeps from the paper
- Spectral comparisons (fast/thermal flux distributions)
- Temperature feedback / void coefficient studies

Follow-on research questions:

- Q1. How sensitive is the 375-day reactivity swing to the online Xe-135 removal rate — can we quantify $dk/d(\lambda_{\text{Xe}})$ and compare to MSRE operational data
- Q2. Extend depletion to 900 days with periodic U-235 additions matching the MSRE operational log and compare drift in k_{eff} to the paper's full-history curve.
- Q3. Swap the fuel to a thorium-bearing MSR composition (LiF-BeF₂-ThF₄-UF₄) and compute the breeding ratio with and without online protactinium separation.
- Q4. Perform a Monte-Carlo-vs-deterministic cross-validation by running an equivalent SCALE/TRITON model on the same 2D slice and quantifying k_{eff} bias as a function of energy-group structure.
- Q5. How does the reactivity evolution change under a 3D axial model with realistic axial power profile and graphite growth (anisotropic swelling) over 2+ years

2.12 12. Physics and Chemistry from Parsimonious Representations: Image Analysis via Invariant Variational Autoencoders (rVAE)

OSTI: 2439897 **Authors:** Ziatdinov et al. (2024) **Domain:** Materials / ML / Microscopy / Generative Models

Coverage: 6/10 **Agreement:** 8/10 **Total:** 14/20

Coverage rationale. Independent PyTorch re-implementation of SE(2)-equivariant rVAE with explicit pose head + coordinate-MLP decoder with random Fourier features, plus a vanilla-VAE baseline. Tested on synthetic hexagonal lattice dataset with one physical factor (lattice constant a) and two nuisances (θ , t_x , t_y). NOT replicated: experimental STEM/SPM microscopy datasets, comparisons against atomai reference, multi-physical-factor datasets, or the broader chemistry/composition case studies from the paper.

Agreement rationale. Central methodological claim reproduced quantitatively: rVAE 2D content latent recovers lattice constant a at Pearson $|r|=0.993$ while pose leakage into z stays ≤ 0.029 ; vanilla VAE with 5D latent only reaches $|r|=0.862$ for a and spreads residual info across pose. rVAE reconstruction MSE 12% lower (79.1 vs 89.3). Only qualitative in that the paper reports visual UMAP panels rather than numerical correlations, so direct number-to-number match is absent.

What reproduced:

- SE(2)-equivariant rVAE encoder (μ_z , $\logvar_z$, θ , t_x , t_y)
- Coordinate-MLP decoder with random Fourier positional features
- Pose applied as inverse coordinate transform on canonical grid
- Linear beta-warmup to avoid posterior collapse
- Content-latent $\rightarrow$ physical-factor disentanglement ($|r|=0.993$ vs 0.862)
- Vanilla VAE baseline entangling content with pose
- Latent-traversal / active-dimension parsimony diagnostic

What missing:

- Original STEM/SPM experimental datasets (proprietary / not linked)
- Comparison against atomai reference implementation
- Multi-physical-factor experiments (composition, strain, polarization)
- UMAP embedding colored by physical parameter (paper’s canonical viz)
- 4D-STEM / defect-class case studies
- Partial rotational symmetry (non-6-fold) datasets

Follow-on research questions:

- Q1. On a synthetic dataset with two coupled physical factors (a AND orientation-angle-of-a-strain-tensor), can the rVAE cleanly disentangle both into separate content dimensions, or does it still recruit only one
- Q2. Benchmark the rVAE against a modern equivariant neural-network baseline (e.g. escnn SE(2)-equivariant CNN autoencoder) on the same hexagonal-lattice task — does explicit equivariance beat the implicit-via-pose-head approach

- Q3. Study the posterior-collapse failure mode systematically: how does the content-to-physics $|r|$ depend on beta schedule shape (linear, exponential, KL-annealing frontend)
- Q4. Apply the rVAE to a real open STEM dataset (e.g. the EMPIAD or NCEM public STEM datasets) and quantify whether the content latent recovers known crystallographic parameters.
- Q5. Extend the pose head to non-trivial symmetry groups (C_4 , C_6 , or reflections) and measure the disentanglement gain on datasets with those specific symmetry groups vs generic $SE(2)$.

2.13 13. FDTD: Solving 1+1D Delay PDE in Parallel

OSTI: 1475143 **Authors:** Zheng, Goldschmidt, Fan (2018) **Domain:** Quantum Optics / Numerical PDE

Coverage: 7/10 **Agreement:** 7/10 **Total:** 14/20

Coverage rationale. Core FDTD 'square rule' solver implemented with $\Delta x = \Delta t$ grid, circular-buffer delay, analytical boundary conditions, and bilinear interpolation for χ construction. Figures 5, 6, 7 reproduced. Figure 8 (non-Markovian regime) not attempted — requires ~125 GB memory for full history. Parallel performance benchmarking (paper's emphasis) not done; serial Python only.

Agreement rationale. Fig 5: zero boundary error, bounded interior error envelope, stable through 40K timesteps (confirmed). Fig 6 ($ka = \pi/4$): $g(0) = 0.001$ at resonance (antibunching), 2.13 off-resonance (bunching) — qualitatively exact. Fig 7 ($ka = \pi/2$): $g(0) = 0.007$ at resonance, 0.539 off-resonance — antibunching reproduced, bunching/oscillation features qualitatively present. No direct reference-solution comparison (paper's Ref [8] scattering theory not implemented).

What reproduced:

- Square-rule FDTD discretization (4-corner averages)
- Stable evolution through 40,000 timesteps (Fig 5)
- Analytical boundary conditions from one-excitation sector $e(t)$ with log-space bounce computation
- $\chi(x, x, t)$ construction with bilinear interpolation
- $g(\tau)$ correlation function (Figs 6, 7)
- Antibunching at resonance ($g(0) \rightarrow 0$) and bunching off-resonance
- Circular-buffer memory management for delay feedback

What missing:

- Figure 8 non-Markovian regime (requires ~125 GB memory)
- Parallel-performance benchmarks (paper's central contribution)
- Reference scattering-theory solution for quantitative g comparison
- Out-of-core or memory-mapped storage for large-grid runs
- Numba/C JIT to realize paper-scale timing
- Pipeline-parallel exploitation of causal delay structure
- Multi-atom / multi-mirror generalizations

Follow-on research questions:

- Q1. Port the solver to CUDA and measure sustained throughput for (N, N) up to the Fig 8 regime — at what grid does the GPU cross the memory wall, and does out-of-core NVMe backing restore feasibility
- Q2. Implement the exact scattering-theory reference (Ref [8]) and quantify FDTD g error as a function of Δ — does the empirical convergence rate match the paper’s theoretical $O(\Delta^2)$
- Q3. Extend the PDE to N atoms at staggered positions — does $g(0)$ show monotonic deepening of antibunching, and is there a critical atom spacing where superradiance flips to subradiance
- Q4. How does $g(0)$ depend on ka beyond $\pi/2$ up to the Fig 8 regime ($ka \approx 10\pi$ – 20π) Produce a phase diagram of $g(0)$ vs $(ka, \omega/)$ showing antibunching/bunching/chaotic zones.
- Q5. Replace the perfect mirror with a finite-reflectivity boundary $r < 1$ — at what r does the non-Markovian oscillation structure collapse to Markovian exponential decay, and does an effective-Markovian approximation work beyond that threshold

2.14 14. Spin-Dependent Scattering of Sub-GeV Dark Matter

OSTI: 3014512 **Authors:** Gori, Knapen, Lin, Munbodh, Suter (2025) **Domain:** Particle Physics / Dark Matter Direct Detection

Coverage: 7/10 **Agreement:** 7/10 **Total:** 14/20

Coverage rationale. Reproduced full rate machinery via DarkELF for three mediator operators (ϕ , a , A') across Al₂O₃ and GaAs, covering DM masses 1 MeV–1 GeV and thresholds 1 meV–1 eV. Reproduced paper Figs. 5–8 (phonon DOS + three mediator reach plots) plus SD vs SI comparison. Missing: constraint plots (Figs 1–4) from SN/SIDM/LHC, Figs 9–10 optimized-mediator-mass scan, experimental exclusions overlays.

Agreement rationale. Reference sigma values for Al₂O₃/ A' /heavy/1meV at $m_\chi \sim 100$ MeV: $1.9 \times 10^{-40} \text{ cm}^2$ (ours) consistent with paper’s Fig. 8 within 10–20% digitization uncertainty. Operator hierarchy ($a < \phi \sim A'$), target hierarchy (Al₂O₃ > GaAs), SD/SI ratio $\sim 10^3$ – 10^4 all reproduced. GaAs normalization off by ~ 5 – $10\times$ vs paper (partial mismatch in nuclear matrix elements). 31/31 unit tests pass.

What reproduced:

- DarkELF multiphonon SD rate computation for ϕ , a , A' operators
- Standard Halo Model parameters ($v_0=220$, $v_e=240$, $v_{esc}=500 \text{ km/s}$, $\rho_\chi=0.4$)
- Figure 5 (Al₂O₃, GaAs phonon partial DOS)
- Figure 6 (ϕ mediator reach) for both targets, heavy/light
- Figure 7 (a mediator) with q^5 integrand behavior
- Figure 8 (A' mediator, $m_{A'} = 10 \text{ GeV}$)
- SD vs SI 10^3 – 10^4 gap
- Isotope spin factors for ²⁷Al, Ga-69/71, As-75

What missing:

- Figures 1-4 (mediator constraints: SN cooling, meson decays, SIDM, LHC)

- Figures 9-10 (optimized mediator mass scan per m_{χ})
- Experimental exclusion overlays (PICO, CRESST, CDEX)
- Massless-mediator SI benchmark
- Full absolute calibration of GaAs rate vs paper (off 5-10x)

Follow-on research questions:

- Q1. How does the SD reach for Al_2O_3 change if realistic experimental resolution ($\sigma_E \sim 0.5$ meV TES) and backgrounds are folded into the 1 meV threshold curves
- Q2. Run the DarkELF computation on a wider target panel (sapphire, GaN, diamond, ZnO, Si_3N_4) at 1 meV threshold and rank candidates by SD reach per kg-yr for each operator.
- Q3. Produce the missing Figs. 9-10 by scanning m_{med} over $[0.1, 10] \cdot q_0$ per m_{χ} and constructing the envelope minimum-sigma curve; quantify how much reach improves vs fixed $m_{\text{med}}=3q_0$.
- Q4. For the A' model, fold in the realistic SN+SIDM constraint band (from the paper's Figs. 1-4) and identify the residual DM-mass window where Al_2O_3 at 1 meV can make a world-leading statement.
- Q5. Compare the DarkELF single-phonon vs multi-phonon+impulse transition by checking the crossover energy and cross-section continuity at $E_{\text{th}} \sim \text{phonon bandwidth}$ for Al_2O_3 .

2.15 15. ScaWL: Scaling k-WL (Weisfeiler-Lehman) Algorithms in Distributed-Memory

OSTI: 2587225 **Authors:** Soss et al. (2022) **Domain:** Graph Algorithms / Parallel & Distributed Computing

Coverage: 6/10 **Agreement:** 7/10 **Total:** 13/20

Coverage rationale. Independent Python/NumPy re-implementation of 2-WL ($k=2$ Folklore Weisfeiler-Lehman) in 210 LOC with multiprocessing-based parallelism. Correctness validated on 3 oracle pairs (iso, C12 vs 2xC6, Shrikhande vs 4x4 rook). Strong-scaling benchmark on single node. NOT replicated: 3-WL (memory-prohibitive in Python), distributed multi-node MPI runs, the UFL Sparse Matrix Collection benchmark suite, the paper's headline 2193x speedup (different absolute baseline).

Agreement rationale. Qualitative strong-scaling curve tracks paper's Table 4 up to physical core count ($p=8$: 6.17x ours vs 7.64x paper). Beyond 10 cores we saturate (hyper-threading limit) while paper's 20-core node reaches 16.06x. Efficiency 0.77 ours vs 0.95 paper at $p=8$ — reasonable given Python multiprocessing overhead vs OpenMP zero-copy. Correctness perfect (3/3 oracles). Problem-size scaling slope ~ 2.3 consistent with $O(n^2 \log n)$ per-iteration cost.

What reproduced:

- 2-WL Folklore refinement algorithm
- Iso / non-iso / strongly-regular correctness oracles
- Strong-scaling curve shape (1..8 processes) matching paper qualitatively
- Termination criterion (color count stable for 2 iterations)
- Rank-reduced hash-key scheme (hashed row/column signatures)

- Problem-size scaling $\sim O(n^2 \log n)$ empirically

What missing:

- 3-WL experiments (memory prohibitive)
- Distributed multi-node MPI runs (single-node Python only)
- UFL Sparse Matrix Collection test suite (used synthetic regular graphs)
- Absolute 2193x speedup vs SparseWL/K-WL baselines
- OpenMP zero-copy shared-memory implementation
- Paper’s storage-complexity claim $O(k|V|^k)$ vs $O(k|V|^{k+1})$

Follow-on research questions:

- Q1. Port the 2-WL implementation to C++/OpenMP on a 20-core node and reproduce the paper’s 16x speedup at $p=20$ on the same synthetic graph — quantify exact Python overhead.
- Q2. Scale to 3-WL on small graphs ($n=40-80$) using the rank-reduction storage trick and compare empirical memory to the paper’s $O(k|V|^k)$ claim.
- Q3. Benchmark 2-WL / 3-WL / 4-WL expressivity on the standard BREC graph-isomorphism benchmark — how many more pairs does $k=3$ vs $k=2$ distinguish in practice
- Q4. Replace deterministic canonicalization with approximate / sketched hashing (e.g. MinHash on neighborhood multisets) and measure speed vs correctness tradeoff on large sparse graphs.
- Q5. Evaluate ScaWL as a subroutine inside a GNN (Provably Powerful Graph Networks) — does distributed k -WL enable larger-graph training and does this improve downstream accuracy on OGB benchmarks

2.16 16. Physics-based hybrid machine learning for critical heat flux prediction with uncertainty quantification

OSTI: 2571909 **Authors:** Furlong, Zhao, Salko, Wu (2025) **Domain:** Thermal hydraulics / ML

Coverage: 6/10 **Agreement:** 7/10 **Total:** 13/20

Coverage rationale. Replicated Biasi-hybrid DNN ensemble vs pure DNN ensemble on both plentiful (80%) and severely limited (9-point) data scenarios using a physics-grounded synthetic surrogate for the NRC CHF database (which is not redistributable). Paper’s Biasi correlation implemented from scratch; five headline metrics reproduced. Not replicated: BNN variant, DGP variant (attempted, blocked by GPyTorch 1.15.2 batch-shape bug), Bowring correlation variant, calibration curves, miscalibration-area analysis, 20-model ensemble (we used 10).

Agreement rationale. Central qualitative finding reproduces unambiguously: Biasi-hybrid $<$ pure DNN $<$ bare Biasi in both data regimes; pure DNN collapses with 9 points (rRMSE 160%, R^2 0.40) while hybrid holds (rRMSE 10.4%, R^2 0.988) — a 15x reduction in error under scarcity. Absolute metric values (our 4.2% mean err vs paper’s 1.85%) differ because our synthetic surrogate carries higher injected noise than the real NRC database; differences are in expected direction and magnitude.

What reproduced:

- Biasi dryout correlation implementation

- Residual-learning hybrid scaffold ($y = y_base + NN(residual)$)
- DNN ensemble (init-diversity)
- Plentiful vs severely-limited scenarios with matched val/test
- Five paper metrics (μ_err , $\max_err$, rRMSE, $F_{>10\%}$, R^2)
- Headline ordering: hybrid > pure ML, esp. under scarcity
- Parity plots for both scenarios

What missing:

- Real NRC CHF experimental database
- Bayesian NN variant
- Deep Gaussian Process variant (blocked by gpytorch 1.15.2 bug)
- Bowring correlation variant
- Calibration curves and miscalibration-area
- Full 20-model ensemble (used 10)
- Operating-regime stratified error analysis

Follow-on research questions:

- Q1. Does the hybrid advantage under data scarcity persist if the base correlation is deliberately mis-specified (e.g., Biasi used outside its valid L/D range) and how quickly does performance degrade
- Q2. For the 9-point regime, can transfer learning from a model pretrained on the synthetic surrogate produce calibrated uncertainty on the real NRC data with <50 fine-tuning points
- Q3. How does epistemic-uncertainty miscalibration area scale with training set size for the three UQ backbones (DNN ensemble vs BNN vs DGP) at matched parameter count
- Q4. Can symbolic regression over the learned residual recover an interpretable correction term to Biasi, and does that term resemble known Groeneveld corrections in the rod-bundle regime
- Q5. When the input-feature set is reduced to operating conditions only (drop geometry), how much of the hybrid advantage disappears, quantifying the contribution of each feature family

2.17 17. Updated Virophage Taxonomy and Distinction from Polinton-like Viruses

OSTI: 2475938 **Authors:** Roux et al. (2023) **Domain:** Virology / bioinformatics

Coverage: 6/10 **Agreement:** 7/10 **Total:** 13/20

Coverage rationale. Reduced reference set (22 NCBI accessions, 13 retained after size filtering) run end-to-end through the paper’s methodological pipeline: Prodigal ORF calling, HMMER scans against the paper’s virophage + PLV HMM profiles, MAFFT alignment + trimAl trimming, IQ-TREE maximum-likelihood phylogenies with 1000 UFBoot replicates on four markers (MCP, ATPase, PRO, penton). Not replicated: full genome set (paper uses hundreds of metagenomic + cultured genomes), ICTV-scale taxonomy assignments, GC-content / gene-content discriminant, deep clade re-classification.

Agreement rationale. Three headline claims reproduce cleanly on the reduced set: (1) HMM classification separates virophages (MCP ≥ 472 bits, four markers positive) from PLV/adintoviruses (zero hits above inclusion threshold) — matches paper’s Fig. 1 / Table 1 binary separation; (2) 4-marker phylogenies are congruent on the backbone (Mavirus basal $\rightarrow$ Sputnik early-diverging $\rightarrow$ SW01+YSLV4 sister pair with 100/100 support in MCP/ATPase/Penton); (3) virophage monophyly is recovered. Paper’s quantitative bit-score ranges match ours qualitatively. Fine-grained family assignments at the ICTV level not attempted.

What reproduced:

- Prodigal ORF prediction
- HMMER3 scan against paper’s 19 virophage + 1 PLV HMMs
- Clean virophage vs PLV/adintovirus separation
- Four-marker MAFFT + trimAl alignments
- IQ-TREE ML phylogenies with 1000 UFBoot
- Backbone topology: Mavirus basal, Sputnik early, SW01+YSLV4 sister
- HMM heatmap and 4-marker congruence figures

What missing:

- Full paper-scale genome collection (hundreds of genomes)
- ICTV-scale family/genus assignments
- GC-content and gene-content discriminant analysis
- Deep-clade (Preplasmiviricetes) re-classification
- Synteny / gene-order analysis across virophages
- Metagenomic-contig based new virophage discovery

Follow-on research questions:

- Q1. Does adding the 32 newly published metagenome-assembled virophage genomes (2023-2024 IMG/VR release) shift the Mavirus-basal topology, or are the paper’s conclusions robust
- Q2. Can a protein language model (ESM-2 or ProtTrans) embedding of MCP and ATPase sequences distinguish virophages from PLVs/adintoviruses without HMM profiles, and at what sequence-identity floor does HMMER fail while LM embeddings still succeed
- Q3. What is the minimum marker-gene subset (MCP alone vs 2/3/4 markers) needed to recover the paper’s family-level topology at ≥ 95 UFBoot support
- Q4. Under Bayesian concordance factor analysis, do the four marker trees support a single evolutionary history or reveal reticulation (horizontal transfer) at the Sputnik/Mavirus split
- Q5. When reference-free clustering (vConTACT3 or mash-based) is applied to the paper’s genome set, does it produce the same family boundaries, or does it over-/under-split relative to the HMM + phylogeny approach

2.18 18. Deep Learning of Atomically Resolved Scanning Transmission Electron Microscopy Images: Chemical Identification and Tracking Local Transformations

OSTI: 1427646 **Authors:** Ziatdinov, Dyck, Maksov et al. (2017) **Domain:** Materials / ML / electron microscopy

Coverage: 6/10 **Agreement:** 7/10 **Total:** 13/20

Coverage rationale. Reproduced the ML pipeline architecture and training protocol: 7.76M-param U-Net (5 encoder stages, BN+ReLU, skip connections, softmax head) in PyTorch 2.5.1 / CUDA 12.1, AdamW + cosine LR, cross-entropy with class weighting, on-the-fly augmentation, 30 epochs on an A100. Paper’s classification target (per-pixel chemical identity map for SrTiO₃ / La_{0.7}Sr_{0.3}MnO₃ [001] perovskite interfaces, 5 classes: Sr, Ti, La/Sr, Mn, vacuum) reproduced. Training data substituted: physics-motivated synthetic surrogate (perovskite lattice + $Z^{1.7}$ HAADF intensity + Poisson noise + scan-line jitter + antisite/vacancy defects) instead of full multislice (Prismatic/abTEM). No evaluation on real experimental STEM frames; no transformation-tracking experiments.

Agreement rationale. On held-out synthetic test set (128 frames, 8.4M pixels): pixel accuracy 0.988, per-class F1 in the 0.949-0.998 range, column-centre RMSE sub-pixel (0.06-0.47 px) for all classes, detection recall ≥ 0.98 for every atom class. These numbers are consistent with the paper’s qualitative claim that an FCN trained purely on simulation produces quantitatively useful chemical mapping, but direct numerical comparison to the paper’s real-STEM metrics is not possible because we never touched an experimental frame. The paper’s transformation-tracking (antisite $\leftrightarrow$ vacancy dynamics over time) was also not attempted.

What reproduced:

- U-Net architecture (5 encoder stages, skip connections, softmax head)
- AdamW + cosine LR + cross-entropy with class weighting
- On-the-fly flip/rotate/brightness augmentation
- Training-from-simulation paradigm
- Per-pixel 5-class perovskite chemical identity
- Sub-pixel column-centre detection (RMSE 0.06-0.47 px)
- Confusion matrix and peak-finding analysis

What missing:

- Full multislice simulation (Prismatic / abTEM) as training data
- Evaluation on real experimental HAADF-STEM frames
- Transformation-tracking (antisite / vacancy dynamics)
- Time-series analysis of local transformations
- Noise / dose-reduction sensitivity sweep
- Generalisation across perovskite chemistries not seen in training

Follow-on research questions:

Q1. When the synthetic training set is replaced with a small-angle-multislice abTEM simulation

(3-4 slices, $\sim 10\times$ cheaper than full Prismatic), does pixel accuracy hold and does generalisation to real STEM frames improve measurably

- Q2. How does the U-Net’s per-class F1 degrade on real experimental HAADF-STEM frames of SrTiO₃/LSMO as dose is reduced from $1e5$ to $1e3$ e-/Å², and at what dose does Sr vs La/Sr confusion become the dominant error
- Q3. Can a self-supervised pretraining stage (masked autoencoder on unlabelled STEM frames) close the synthetic-to-real domain gap with $<5\%$ real-frame labels
- Q4. Does replacing the U-Net with a Vision Transformer + UperNet head preserve sub-pixel detection precision while providing calibrated classification uncertainty per atom column
- Q5. When the model is applied across a temporal sequence of experimental STEM frames, does a simple Kalman smoother on per-column class probabilities recover transformation trajectories (antisite $\leftrightarrow$ vacancy) with known drift statistics

2.19 19. Simple Coplanar Waveguide Resonator Mask Targeting Multiplexed Superconducting-Qubit Readout

OSTI: 1983793 **Authors:** (Argonne CPW mask paper) (2022) **Domain:** Quantum Devices / Superconducting circuits

Coverage: 5/10 **Agreement:** 7/10 **Total:** 12/20

Coverage rationale. Analytical results replicated: CPW impedance via conformal mapping ($48.3\ \Omega$ vs $50\ \Omega$ target), 8-resonator frequency allocation 4.6–7.4 GHz, quarter-wave lengths, Q_c vs coupler length. Participation ratios computed with 2D finite-difference Laplace solver (278K-point non-uniform grid, 5 nm near interfaces). Missing: paper’s full 3D FEM eigenmode + driven simulations (Ansys HFSS or Palace), full mask-layout simulation with GDS, adaptive sub-nm meshing, experimental Q_i measurements.

Agreement rationale. Analytical values excellent: Z within 3.4%, C within 14%, frequencies exact. Participation ratios: $p_{MS}/p_{SA} = 2.09$ vs paper 2.00 (excellent). Absolute p values within $\sim 2\times$ due to 2D thin-layer approximation. Substrate participation 91.85% vs 89% (3% error). p_{MA} overestimated $6\times$ — known limitation of 2D solver near conductor edge singularity. Q_c vs coupler length tunable over 3+ decades (matches Fig 3).

What reproduced:

- Conformal-mapping CPW impedance formula ($Z = 48.3\ \Omega$, $\epsilon_{eff} = 6.225$)
- 8-resonator frequency plan (4.6, 5.0, 5.4, 5.8, 6.2, 6.6, 7.0, 7.4 GHz)
- Quarter-wave resonator lengths (4.06–6.54 mm)
- Participation ratio ordering $p_{MS} > p_{SA} > p_{MA}$
- Q_c vs coupler length monotonic decrease over 3+ decades
- p_{MS}/p_{SA} ratio = 2.09 (paper: 2.00)
- 2D FD Laplace solver with non-uniform grid

What missing:

- Full 3D FEM eigenmode simulation (Ansys HFSS or Palace)
- Driven S-parameter simulation of the 8-resonator-on-feedline chip

- Sub-nm adaptive meshing to resolve 3 nm interface layers accurately
- GDS mask-level layout import and simulation
- Kinetic-inductance correction to Z
- Experimental Q_i measurements and TLS loss-budget fitting
- Realistic isotropic-etch trench geometry (curved profile)

Follow-on research questions:

- Q1. Run the same participation analysis in Palace (AWS open-source FEM) on the published GDS mask — do the absolute p_{MS} and p_{MA} values drop toward the paper’s $\times 10^3$ scale when sub-nm interface meshing is used
- Q2. Sweep trench depth 0–300 nm in 10 nm steps — does p_{SA} decrease monotonically, and where is the diminishing-returns knee for TLS loss reduction
- Q3. How sensitive is $p_{MS}/p_{SA} \approx 2$ to center-conductor-to-gap aspect ratio s/g Sweep $s \in \{3,6,9,12\}$ m holding $Z=50 \Omega$ via g adjustment; report the ratio’s $\pm 3\sigma$ band across realistic fab tolerances.
- Q4. Model kinetic inductance of a 100 nm Nb film with $\lambda_L = 80$ nm and re-compute Z and f — does the 3.4% gap close, and do all 8 frequencies shift consistently
- Q5. Couple the analytical $Q_c(c)$ curve to the participation-ratio loss model — can a single scalar ‘design efficiency’ metric recommend an optimal coupler length that jointly minimizes TLS loss and maximizes multiplexed readout fidelity

2.20 20. Electronic and Optical Properties of Two-Dimensional GaN from First-Principles

OSTI: 1484740 **Authors:** Bayerl, Islam, Jones, Protasenko, Jena, Kioupakis (2017) **Domain:** Condensed Matter / 2D Materials DFT-GW-BSE

Coverage: 5/10 **Agreement:** 7/10 **Total:** 12/20

Coverage rationale. Replicated DFT-level monolayer results: structural relaxation ($a=3.219$ Å PBE vs 3.165 Å LDA), direct band gap at Gamma (2.99 eV vs 2.95 eV), band structure along Gamma-M-K-Gamma, biaxial strain curve (-5% to +5%). Partial bilayer (convergence issues; gap 0.065 vs 1.32 eV). NOT replicated: GW quasiparticle corrections (+3.37 eV), BSE exciton binding (1.31 eV), optical absorption, radiative lifetimes, internal E-field 74 MV/cm — i.e. the paper’s many-body contributions.

Agreement rationale. Monolayer DFT gap within 1.4% of paper (2.99 vs 2.95 eV). Lattice constant brackets experiment (3.219 Å PBE vs 3.165 Å LDA vs 3.190 Å expt), consistent with PBE-vs-LDA systematics. Strain-gap curve shape and asymmetry faithfully reproduced. Bilayer badly off (0.065 vs 1.32 eV) due to incomplete relaxation + PBE/LDA differences for polar slabs. Direct-gap character, valence-band quasi-degeneracy, and conduction-band dispersion all reproduced.

What reproduced:

- Monolayer H-passivated structure and buckling
- DFT band gap 2.99 eV (PBE) vs paper 2.95 eV (LDA)
- Direct gap at Gamma with N 2p valence-top character

- Band structure topology along Gamma-M-K-Gamma
- Biaxial strain dependence of gap (asymmetric, peak near 0-1%)
- SG15 ONCVSP pseudopotentials with Ga 3d in valence

What missing:

- GW quasiparticle corrections (+3.37 eV monolayer, +2.68 eV bilayer)
- BSE exciton binding 1.31 eV (monolayer), 0.75 eV (bilayer)
- Optical absorption spectra / optical gap 5.01 eV
- Radiative lifetimes (0.6 ns monolayer, 4.6 ns bilayer)
- Internal polarization field 74 MV/cm (monolayer), 65 (bilayer)
- Bilayer DFT gap (off by ~1.25 eV)
- Singlet-triplet splitting (63 vs 6 meV)

Follow-on research questions:

- Q1. Run GW@PBE on the existing Quantum ESPRESSO starting point using BerkeleyGW on a single Spark / small HPC slice — how close is G0W0 monolayer gap to the paper's 6.32 eV
- Q2. Redo the bilayer structural relaxation with tighter force tolerance ($<1e-4$ Ry/Bohr) and a larger vacuum + dipole correction; does the DFT gap recover toward 1.3 eV
- Q3. Compute the biaxial strain dependence of the GW-corrected band gap (via scissor-proxy: assume GW correction is strain-independent) and compare to strain-dependent optical gap in GaN/AlN heterostructure experiments.
- Q4. Replace H passivation with F or Cl and track how the band gap, polarization field, and exciton binding change — identify a passivation that turns the bilayer direct-gap.
- Q5. Use a machine-learned xc functional (e.g. SCAN, r2SCAN, or DM21) as a PBE alternative and quantify the DFT-functional uncertainty envelope on the 2D GaN gap before GW corrections.

2.21 21. Numerical study of the ignition behavior of a post-discharge kernel in a turbulent stratified crossflow

OSTI: 1559043 **Authors:** Jaravel, Labahn, Sforzo, Seitzman, Ihme (2019) **Domain:** Combustion / LES

Coverage: 6/10 **Agreement:** 6/10 **Total:** 12/20

Coverage rationale. Three replication generations attempted. (1) CPU simplified (v1) scored 2/5 — kernel collapsed to a sub-cell nub due to an SI/CGS units bug in prob_parm, peak T = 1673 K instead of 3300 K. (2) PeleLMeX GPU port abandoned because the low-Mach formulation is not physically compatible with a hot plasma-kernel initial condition. (3) PeleC v2 (scored here) fixed the units bug, correctly initialised a 2 mm radical-seeded kernel straddling the fuel/air splitter at 3298 K, ran a 4-phi sweep (0.6/0.8/1.0/1.2) on 4x A100 to t~65 us with DRM19 chemistry. Not replicated: sustained thermal runaway (window too short), ensemble of 5 realisations per phi for the ignition-probability curve (still running on Polaris/chiatto00 at report time), AMR + full inflow turbulence generator.

Agreement rationale. Observed clean monotonic ordering by phi of peak OH (0.0131 -> 0.0187),

H₂O (0.053 → 0.093), and CO₂ (0.043 → 0.075) mass fractions — the qualitative precursor to the paper’s central ignition-probability-vs-phi result. However we do not produce a probability curve (single realisation per phi, not 5), and thermal runaway was not observed within the ~65 us simulated window (T_{max} seed-dominated, decays monotonically ~72 K). Mechanism-level agreement is strong; quantitative ignition-probability comparison is out of reach until the Polaris ensemble completes.

What reproduced:

- Compressible LES deployment (PeleC + AMReX) stable for >1000 steps on 4x A100
- Correct 3298 K radical-seeded kernel at fuel/air splitter interface
- DRM19 methane-air chemistry integration
- Four-phi sweep (0.6, 0.8, 1.0, 1.2) at matched initial kernel state
- Monotonic phi-ordering of peak OH, H₂O, CO₂ mass fractions
- SI/CGS units-bug diagnosis and fix (documented root cause)
- Time-series and phi-trend figures

What missing:

- Sustained thermal runaway / self-sustained flame propagation
- Ensemble statistics (5 realisations per phi — running on Polaris)
- Paper’s ignition-probability curve vs phi (requires ensemble)
- AMR activation
- Full synthetic-turbulence inflow generator
- Extension to t ~ 300-500 us to capture ignition delay
- v1 CPU and PeleLMEx GPU attempts (documented as failures/incompatibilities)

Follow-on research questions:

- Q1. When the simulated window is extended to t = 500 us with AMR active (refinement around OH > 1e-3 isocontour), does the single-realisation case at phi = 1.0 transition to self-sustained propagation, and what is the measured ignition delay
- Q2. How does the monotonic OH-vs-phi ordering change if the kernel radical composition (Y_{OH} = 0.01, Y_O = 0.01, Y_H = 0.001) is replaced with an equilibrium-plasma mixture at 3300 K, and does ignition probability become phi-insensitive
- Q3. Can a low-cost analytic surrogate (kernel heat release + local Da number) predict the paper’s ignition-probability curve from 2-3 PeleC realisations plus a calibrated stochastic model, reducing the required ensemble size by 5x
- Q4. What is the sensitivity of the phi-ordering to the DRM19 → GRI-3.0 chemistry upgrade, and does the additional NO_x chemistry change the OH pool saturation plateau
- Q5. Under cross-flow velocity scaling (10-40 m/s), does the critical phi for ignition shift monotonically with the local Da number, and can the paper’s single-velocity result be collapsed onto a Da-based master curve

2.22 22. Generative AI-Driven Accelerated Discovery of Passivation Molecules for Perovskite Solar Cells

OSTI: PVMol-Gen-Fajar2026 **Authors:** Fajar et al. (2026) **Domain:** Materials/Generative ML (perovskite passivation)

Coverage: 7/10 **Agreement:** 5/10 **Total:** 12/20

Coverage rationale. Full three-stage pipeline implemented (SMILES-X classifier, iterative GPT-2 generation for 3 cycles at paper scale of 100K/cycle, 7 RDKit filters + clustering). Ran on real 8×A100 hardware. Missing: xTB/DFT E_{gap} and dipole filtering (used RDKit surrogates), and paper’s exact SMILES tokenization pathway (we used SELFIES variant in production run).

Agreement rationale. Generation rates reasonable (83-88% class-1 per cycle). Key quantitative gap: Stage-1 classifier F1=0.656±0.031 vs paper 0.80, ROC-AUC 0.620 vs 0.88 — major reproducibility concern pointing to library/hyperparameter differences we could not pin down. SELFIES pathway yields 6.6× more filtered candidates than SMILES, so downstream numbers diverge by design. Final 10-cluster representatives produced but not externally validated.

What reproduced:

- SMILES-X-style discriminative classifier with 5-fold CV on 314 labeled molecules
- PubChem similarity expansion ($\geq 80\%$ Tanimoto) to build ~11K training set
- GPT-2 fine-tuning + generate-classify-retrain loop for 3 cycles at 100K molecules/cycle
- Stage-3 RDKit filters (SA score, PAINS, HBD/HBA, TPSA)
- Clustering into 10 groups with representative selection
- Overall pipeline architecture end-to-end at paper scale

What missing:

- Matching classifier metrics (F1 0.656 vs 0.80) — root cause not identified
- xTB/DFT energy gap and dipole filters (used RDKit approximations)
- Paper’s exact SMILES augmentation protocol (we swapped to SELFIES mid-pipeline)
- External experimental validation of top candidates
- Synthetic accessibility tie-breaking rules applied by paper for final down-select
- DFT-level verification of E_{gap}∈[1.5,5.0] eV and dipole∈[1.5,4.0] D per molecule

Follow-on research questions:

- Q1. Does matching the paper’s exact SMILES-X version + tokenizer close the F1 gap from 0.656 to 0.80, or is the gap due to training-data preprocessing (canonicalization, augmentation multiplier)
- Q2. Run xTB on the 10 final candidates from both SMILES and SELFIES pipelines — do the E_{gap}/dipole distributions overlap with the paper’s reported 10 molecules
- Q3. Does training-set contamination from predicted class-1 relabeling (cycles 2-3) inflate apparent classifier accuracy Quantify by holding out a fixed external test set and tracking its F1 across cycles.
- Q4. Substitute the GPT-2 generator with a chemistry-tuned foundation model (e.g., ChemGPT,

MolGPT) — does the class-1 rate at cycle 1 jump from 83% to >95%, and do top-10 clusters converge to similar motifs

Q5. Benchmark downstream perovskite figure-of-merit predictors (DFT binding energy on Cs/FA surfaces) on the 20 candidates (10 SMILES + 10 SELFIES) to test whether the representation choice biases toward different chemistries.

2.23 23. A Portfolio Approach to Massively Parallel Bayesian Optimization

OSTI: 2571540 **Authors:** Binois, Collier, Ozik, Wozniak (2021) **Domain:** ML / Bayesian Optimization

Coverage: 5/10 **Agreement:** 6/10 **Total:** 11/20

Coverage rationale. Noiseless Tier-1 benchmarks replicated: Branin-2D/12D and Hartmann6/Hartmann6-12D with qHSRI, qEI, qTS, PF, RS. Reduced budget (n=200 vs 500), 10 seeds vs 20, q=10 only (no scaling). Missing: noisy benchmarks (paper’s strongest selling point, with replication allocation), multi-objective benchmarks, application benchmarks (CNN/CityCOVID/Lunar Lander), MBO and qAEI baselines, scaling q=25..500. Python/BoTorch reimplementation of R/DiceOptim/GPareto pipeline.

Agreement rationale. Qualitative ranking correct: qHSRI and PF are top non-EI methods across all four benchmarks; qEI catastrophically underperforms (often worse than RS), matching paper’s claim. Absolute optimality gaps larger than paper (e.g., 47.5 vs 1-3 on Branin-12D) because of reduced budget and different n_init. Unexpected qTS dominance in our runs not emphasized in paper — likely BoTorch posterior-sampling artifact. qEI failure is more severe in our BoTorch implementation than in the paper’s R code.

What reproduced:

- qHSRI Sharpe-ratio portfolio selection with SLSQP QP solver
- Hypervolume indicator computation in (mean, std) space
- 4 noiseless benchmarks: Branin-2D, Hartmann6, Branin-12D, Hartmann6-12D
- Baselines: qEI, qTS, PF, RS
- Core claim: portfolio methods (qHSRI, PF) > standard qEI in batch BO
- GP model with Matérn 5/2 kernel and exact marginal-likelihood fitting

What missing:

- Noisy benchmarks with replication allocation (Figs 3, 9)
- Multi-objective benchmarks (Figs 4, 10)
- Application benchmarks: CNN, CityCOVID, Lunar Lander
- MBO and qAEI baselines (R-only)
- Scaling experiments at q=25, 50, 100, 200, 500
- Full n=500 budget and 20 seeds for variance tightening
- Timing comparison and absolute wall-clock scaling

Follow-on research questions:

- Q1. Reproduce the noisy Branin/Hartmann6 benchmarks in Python with heteroscedastic GP — does qHSRI’s automatic replication allocation match the paper’s claimed advantage over qAEI
- Q2. Why does BoTorch’s qEI underperform so dramatically (worse than RS) Instrument the joint-acquisition optimization and diagnose: is it L-BFGS local minima, the raw-samples heuristic, or kernel conditioning
- Q3. Test qHSRI at $q=100$ on a 50D Ackley benchmark to verify the paper’s $O(1)$ batch-cost claim — does wall-clock per iteration stay flat vs q , and does regret scale as $\sqrt{\log q}$
- Q4. Benchmark qHSRI against modern batch methods released after 2021 (TurBO, LaMBO, BORE) on the same 4 noiseless benchmarks with identical n , q , seeds — where does the portfolio approach sit in the 2026 landscape
- Q5. Combine Thompson sampling candidates (our surprisingly strong qTS) with Sharpe-ratio selection in a hybrid qTS-HSRI — does the combined method beat both on Hartmann6-12D

2.24 24. Variational Monte Carlo calculations of $A \leq 4$ nuclei with an artificial neural-network correlator ansatz

OSTI: 1864334 **Authors:** Lovato, Adams, Carleo, Rocco et al. (2022) **Domain:** Nuclear physics / ML

Coverage: 5/10 **Agreement:** 6/10 **Total:** 11/20

Coverage rationale. Reproduced the method skeleton (PyTorch NN correlator ansatz + Metropolis MC + gradient-descent optimisation + local energy via autodiff Laplacian) on $A=2$ and $A=4$, but substituted the Minnesota central NN potential for AV6’/AV8’, used Adam instead of stochastic reconfiguration, and employed a purely-spatial symmetric ansatz without explicit tensor/spin-orbit/backflow/three-body correlators. $A=3$ nuclei (^3H , ^3He) not attempted.

Agreement rationale. Deuteron $E_0 = -2.200 \pm 0.001$ MeV reproduces Minnesota-triplet exact (-2.202) to 0.1% and experimental (-2.2246) to 1.1% — excellent. ^4He $E_0 = -24.13 \pm 0.10$ MeV is a legitimate variational upper bound but sits 15% above experimental -28.30 MeV and 19% above Minnesota Faddeev-Yakubovsky -29.94 MeV, worse than the paper’s $\sim 1\text{-}3\%$ agreement with GFMC benchmarks. Qualitative methodology (convergence, near-Gaussian local-energy distribution) reproduces cleanly; quantitative $A=4$ accuracy falls short because operator correlators and backflow were omitted.

What reproduced:

- GPU PyTorch NN-correlator VMC code (~ 300 LOC)
- Metropolis sampling with adaptive step size
- Autodiff kinetic energy via double-backward Laplacian
- Deuteron binding energy to 1.1% of experiment
- ^4He variational upper bound that respects $E_V \geq E_0$
- Quantile-clipped gradient training with $\sim 50\%$ acceptance

What missing:

- AV6’/AV8’ operator-dependent NN potential (used Minnesota central only)
- Stochastic-reconfiguration / natural-gradient optimiser (used Adam)

- Explicit tensor, spin-orbit, isospin, backflow, three-body correlators
- A=3 nuclei (3H and 3He)
- Agreement with GFMC benchmarks to ~1-3% for A=4
- Full spin-isospin walker statistics (handled analytically)

Follow-on research questions:

- Q1. Does adding a differentiable backflow transformation (learned per-pair coordinate shifts) to the symmetric ansatz close most of the 6 MeV Minnesota-central 4He gap, and how does the closure scale with ansatz width
- Q2. For A=4 under Minnesota, how does the stochastic-reconfiguration optimiser compare to Adam in (a) final energy and (b) sample efficiency at matched wall time on a single A100
- Q3. Can a permutation-equivariant transformer correlator generalise across A=2..4 in a single training run (transfer-learning the pair correlator with additional three-body heads)
- Q4. What is the minimum network capacity (parameters, depth, width) at which the deuteron accuracy saturates at the exact Minnesota bound, and does that minimum scale super-linearly with A
- Q5. When the Minnesota potential is replaced by a spin-orbit-augmented version (Minnesota + LS), does the NN correlator recover the spin-orbit-induced tensor correlations, or must they be supplied via an operator ansatz

2.25 25. Effect of Single Atom Platinum (Pt) Doping and Facet Dependent on the Electronic Structure and Light Absorption of Lanthanum Titanium Oxide (La₂Ti₂O₇): A Density Functional Theory Study

OSTI: 1981773 **Authors:** et al. (LTO+Pt DFT) (2022) **Domain:** Materials / DFT
Coverage: 5/10 **Agreement:** 6/10 **Total:** 11/20

Coverage rationale. Reproduced the bulk La₂Ti₂O₇ PBE single-point, the clean (001) slab (44 atoms), and the Pt-doped (001) slab with one surface Ti -> Pt substitution, including SCF + NSCF electronic structure and total DOS — using open-source Quantum ESPRESSO 7.4.1 GPU (paper used VASP). Not replicated: second (010) facet (88-atom slab was prepared but not completed), frequency-dependent optical absorption $\alpha(\omega)$, Bader charge analysis, spin-polarised calculation on the Pt-doped slab. Orbital-resolved PDOS obtained only after re-running NSCF with Gaussian smearing (tetrahedra_opt + projwfc.x bug in this QE build).

Agreement rationale. Qualitative principal finding of the paper reproduces: single-atom Pt substitution narrows the (001) effective band gap dramatically (2.31 eV clean -> 0.55 eV Pt-doped, a 1.77 eV / 76% reduction), with CBM drop 1.52 eV and VBM rise 0.24 eV consistent with Pt 5d hybridisation with Ti 3d and O 2p. Bulk lattice parameters agree to <1% of experiment. Bulk PBE gap (2.87 eV) is lower than paper's ~3.8 eV (code/pseudopotential/cutoff differences); both underestimate experiment as expected for GGA. Single-facet only, so paper's facet-dependence claim cannot be confirmed in full.

What reproduced:

- Bulk La₂Ti₂O₇ PBE DFT with SSSP Efficiency v1.3.0 pseudopotentials
- Lattice parameters to <1% of experiment

- (001) symmetric slab (44 atoms, 15 Å vacuum, bottom 14 fixed)
- Pt substitution at surface Ti site (one atom)
- SCF and NSCF electronic structure for both slabs
- Band-gap narrowing: 2.31 eV $\rightarrow$ 0.55 eV on Pt doping
- Total DOS and (eventually) orbital-resolved PDOS

What missing:

- Second facet (010) comparison — paper’s facet-dependence claim
- Frequency-dependent optical absorption $\alpha(\omega)$ via `epsilon.x` / `turbo_lanczos`
- Bader charge analysis
- Spin-polarised calculation of Pt-doped slab
- Work-function and surface-dipole analysis
- Full VASP-equivalent PBE gap (our 2.87 vs paper’s $\sim$ 3.8 eV)
- vc-relax on bulk cell

Follow-on research questions:

- Q1. What is the facet-dependence of the Pt-induced band-gap narrowing when computed on (010) and (111) surfaces with matched slab thickness, and does (001) remain the most photoactive as the paper claims
- Q2. Does a spin-polarised calculation of the Pt-doped (001) slab reveal a local magnetic moment on Pt, and does that shift the mid-gap states relative to the non-spin-polarised result
- Q3. How does the gap narrowing evolve with Pt coverage (1/8, 1/4, 1/2, 1 ML) on the (001) surface, and at what coverage does the system become metallic
- Q4. Using HSE06 or G0W0 on the relaxed Pt-doped slab, does the quantitative gap narrowing (absolute gap and CBM/VBM shifts) survive, or is it a PBE self-interaction artefact
- Q5. Can a climbing-image NEB calculation of water-splitting intermediates (H^* / OH^*) on the Pt-doped (001) surface connect the electronic-structure change to a measurable overpotential difference vs the clean surface

2.26 26. Latent Stochastic Differential Equations for Modeling Quasar Variability and Inferring Black Hole Properties

OSTI: 2396968 **Authors:** Fagin et al. (2024) **Domain:** Astrophysics / ML

Coverage: 4/10 **Agreement:** 6/10 **Total:** 10/20

Coverage rationale. Replicated a single-band simplification of the paper’s architecture: encoder (GRU) $\rightarrow$ latent Ito SDE (learned prior drift, diagonal diffusion) $\rightarrow$ decoder, trained with reconstruction NLL + Girsanov KL via `torchsde.sdeint(..., logqp=True)`. Did NOT replicate the 6-band LSST multivariate structure, the Sim5 GR ray-tracing physical accretion-disk transfer functions, the 9-parameter Cholesky-Gaussian head for black-hole parameter inference, the 100k-curves/epoch regenerated training volume, or the full LSST opsim cadence. Training set was $\sim$ 2000x smaller, model $\sim$ 17x smaller, compute $\sim$ 3000x less (160 s CPU vs paper’s $\sim$ 6 weeks V100).

Agreement rationale. Methodology reproduces correctly where implemented. Quantitative: our single-band latent-SDE RMSE 0.198 mag vs paper’s 6-band latent-SDE RMSE 0.0959 mag — direct comparison not apt given the single-band simplification (most of the paper’s edge over GPR comes from sharing information across six bands). In the 1-band limit our GPR baseline (Matern-1/2, LOO) beats the latent-SDE (RMSE 0.048 vs 0.198), which is expected because Matern-1/2 IS the DRW covariance. Trends and qualitative behaviour match; the paper’s central contributions (physical inference, multiband) are untouched.

What reproduced:

- Encoder (GRU) -> Latent Ito SDE -> MLP decoder architecture
- Girsanov KL via torchsde.sdeint(logqp=True)
- Adam training with linear KL annealing
- DRW driving-signal simulation
- Single-band reconstruction (mag-space RMSE/MAE/NLL)
- GPR Matern-1/2 baseline comparison

What missing:

- 6-band LSST multivariate architecture
- Sim5 GR ray-tracing transfer functions
- 9-parameter Cholesky-Gaussian physical-parameter head
- 100k curves/epoch regenerated training volume
- Full LSST opsim cadence
- 6-week V100 training-compute regime
- Posterior calibration diagnostics

Follow-on research questions:

- Q1. How does the latent SDE’s parameter recovery degrade as photometric noise scales to 10x LSST requirement
- Q2. Can replacing the GRU encoder with a Transformer (time-series foundation model like TimesFM) improve parameter recovery for the weak-signal parameters (a, lambda_Edd)
- Q3. What is the minimum survey duration (vs LSST 10yr) needed to recover log(M_BH) with 0.1 dex precision
- Q4. Does adding broad-line region emission (reverberation mapping light curves) as an auxiliary input tighten black hole mass constraints
- Q5. How robust is the latent SDE posterior calibration under model mismatch (e.g., the accretion disk is not a simple thin-disk lamppost)

2.27 27. Cosmic Reionization On Computers: Properties of the Post-Reionization IGM

OSTI: 1275503 **Authors:** Gnedin, Becker, Fan (2017) **Domain:** Cosmology / Reionization / IGM

Coverage: 5/10 Agreement: 5/10 Total: 10/20

Coverage rationale. Attempted all five GBF17 statistical analyses (flux PDF, dark gaps, gap $\tau_{\min}$ sensitivity, transmission peaks, DC mode + reionization history) but using a 1D log-normal FGPA forward model rather than the full CROC AMR+OTVET radiation-hydrodynamic simulation. 500 sightlines x 5 z-bins computed in ~ 3 min vs paper's 10^4 -core months of compute. Major simplification of the underlying physics.

Agreement rationale. Qualitative trends reproduced across all five probes (flux-PDF shape, gap morphology, $\tau_{\min}$ sensitivity, missing-wide-peaks, negative DC-mode correlation). Quantitatively: τ_{eff} 30-60% too high at $z < 5.5$ and factor-of-several too opaque at $z > 5.9$. Same $5.5 < z < 5.7$ simulation-vs-data tension as CROC, but amplified in our model. Shape is right; absolute calibration fails above $z = 5.5$ — fairly honest outcome for a semi-analytic stand-in.

What reproduced:

- Flux PDF cumulative shape in 5 z-bins
- Dark gap length distributions for $\tau_{\min} = 2.5, 3.0, 3.5$
- $\tau_{\min}$ sensitivity: shape preserved, tail shifts
- Transmission peak heights ($h_p \sim 0.05$) but narrow widths
- Lack of wide peaks (attributed to missing QSO proximity zones)
- DC-mode negative correlation $r(\delta, \tau_{\text{eff}}) \sim -0.06$
- Sigmoid reionization history $z_{\text{re}} \sim 7$, complete at $z \sim 5.5$

What missing:

- Full 3D AMR hydrodynamics (ART code) with 100 pc resolution
- Self-consistent OTVET radiative transfer
- Non-equilibrium chemistry + temperature evolution
- 3 realizations x 1024^3 vs our 500 1D skewers
- Quantitative agreement at $z > 5.5$ (factor-of-several off)
- Individual quasar proximity zones
- HeII reionization heating

Follow-on research questions:

- Q1. Can a learned correction to the FGPA lognormal tail (e.g. a small neural network trained to map 1D lognormal $\rightarrow$ CROC flux PDF) close the $z > 5.5$ gap while retaining the 10^4 -fold speedup
- Q2. Inject a toy proximity-zone population (Poisson-distributed bright QSOs with PDF-convolved ionization bubbles) and test whether the missing wide-transmission-peak population is explained.
- Q3. Re-calibrate the UV-background fluctuation field (σ_{Γ} , ℓ_{mfp}) by fitting simultaneously to flux PDF + dark-gap statistics at $z < 5.5$ and check whether the fit is unique or degenerate.
- Q4. Extend the semi-analytic model down to $z = 4$ and up to $z = 7$ and compare against the paper's

full redshift range — does the disagreement scale with the mean transmitted flux

- Q5. Use the DC-mode test to derive an effective $b(z, \delta)$ bias of the Lyman-alpha forest in the reionization epoch and compare to theoretical predictions from the patchy-reionization literature.

2.28 28. Learning Sequential Distribution System Restoration via Graph-Reinforcement Learning

OSTI: 1868518 **Authors:** Zhao, Wang (2022) **Domain:** Power Systems / Deep Reinforcement Learning

Coverage: 5/10 **Agreement:** 5/10 **Total:** 10/20

Coverage rationale. Independently implemented a GCN + Dueling Double DQN multi-agent architecture with parameter sharing plus MLP-DQN and heuristic baselines on IEEE 33-bus and procedurally-generated IEEE 123-bus networks using pandapower. NOT replicated: the IEEE 8500-node scalability experiment, the CPLEX MILP comparison, and pre-training/transfer-learning (123->8500) paradigm. The paper is behind paywall and no public code, so methodology was reconstructed from abstract + survey literature.

Agreement rationale. Qualitative matches: GCN-DQN > random/MLP baselines on 123-bus, sequential restoration improves over training, GCN inference ~linear in $|V|$. Quantitatively GCN-DQN and MLP-DQN perform comparably on the 33-bus (small network), and GCN marginally better on 123-bus (0.141 peak vs comparable MLP). No direct numeric curve from paper is reproduced — only structural claims. NaN-loss issues during training suggest the replica’s hyperparameters are not tuned to paper-quality.

What reproduced:

- GCN + Double DQN + Dueling multi-agent architecture
- Parameter sharing across DG agents
- pandapower-based restoration MDP with switches + DGs
- IEEE 33-bus and 123-bus test feeders
- GCN-DQN outperforming random and (on 123-bus) MLP
- Sequential load recovery over episodes
- Inference time ~linear in bus count

What missing:

- IEEE 8500-node scalability (the headline scale claim)
- Pre-training on 123-node + fine-tuning on 8500-node (transfer learning)
- CPLEX MILP comparison on large networks
- Exact paper reward weighting and bonus scheme
- Standard IEEE 123 switch topology (we procedurally generated ours)
- Robustness / ablation studies from paper
- GPU-scale training

Follow-on research questions:

- Q1. Run the trained GCN-DQN on a full IEEE 8500-node case (e.g. from OpenDSS test feeders) and measure zero-shot transfer vs fine-tuned performance — does the GCN’s topology-aware embedding genuinely generalize
- Q2. Compare GCN-DQN against modern graph-transformer policies (e.g. Graphormer, GPS) with identical training budgets to test whether the paper’s GCN advantage holds under stronger architectural baselines.
- Q3. Formulate the restoration problem as MILP in pandapower+pyomo and quantify the optimality gap of GCN-DQN — how close does RL get to CPLEX-optimal on networks where MILP is tractable
- Q4. Add adversarial fault scenarios (targeted attacks rather than random 10-30% line outages) and measure GCN-DQN robustness vs the greedy heuristic baseline.
- Q5. Study reward-shaping alternatives: compare the paper’s weighted load/voltage/thermal reward against a strict load-restoration-only reward and a constraint-violation-barrier reward — which gives the most reliable policy in realistic fault scenarios

2.29 29. Divide and Conquer: Learning Chaotic Dynamical Systems with Multi-Step Penalty Neural ODEs (MP-NODE)

OSTI: 3003857 **Authors:** Chakraborty, Chung, Arcomano, Maulik (2024) **Domain:** Scientific ML / Chaotic Dynamical Systems

Coverage: 5/10 **Agreement:** 4/10 **Total:** 9/20

Coverage rationale. Reproduced Section 4.1.1 (gradient explosion), 4.1.2 (loss landscape, partial), Lorenz-63 vanilla vs MP-NODE, and attempted KS equation. NOT replicated: Kolmogorov flow (Section 4.3), ERA5 climate emulation (Section 4.4), which are the paper’s headline high-dimensional results. Used PyTorch + RK4 vs paper’s likely JAX + Tsit5; architecture choices (Lorenz MLP, KS dilated-CNN) partially reconstructed.

Agreement rationale. Gradient reduction qualitatively reproduced but quantitatively much smaller ($\sim 40\times$ vs paper’s $\sim 10^8\times$) because $T=10$ vs paper’s $T=20$ is too short for full chaotic amplification. MP-NODE beats vanilla on Lorenz training loss (22.5 vs 81.2). Attractor statistics mismatch: our MP mean $z=97$ vs true 23.5 (neither model fully converged). KS experiment collapsed to zero output — effectively failed. So MP > vanilla qualitative claim holds; quantitative claims only partially.

What reproduced:

- Multi-step penalty augmented loss formulation (Algorithm 1, Eqs. 16-24)
- Gradient-magnitude reduction of MP vs vanilla (qualitative)
- Similar computational cost (MP $\sim 5\%$ overhead)
- MP-NODE lower L_{GT} vs vanilla on Lorenz-63 (3.6x lower)
- Multi-step window + μ -scheduling machinery
- Vanilla NODE attractor collapse vs MP-NODE retaining orbital structure

What missing:

- Section 4.3 Kolmogorov flow (2D NS turbulence)
- Section 4.4 ERA5 atmospheric emulation
- Quantitative gradient explosion 10^8 factor (hardware-limited)
- Working KS joint-PDF reproduction (architecture collapse)
- Paper Table 1 KL divergence 0.029 for MP-NODE on KS
- Original JAX implementation; stabilized CNN-NODE of Linot 2023
- Rigorous loss-landscape non-convexity demonstration

Follow-on research questions:

- Q1. Rerun the gradient explosion demo at $T=20$ and 2000 steps (matching paper) on a single A100 / Spark to confirm the 10^8 x gap — does MP’s advantage scale with T as $\exp(\lambda_1 * T)$
- Q2. Implement Linot-2023’s stabilized NODE architecture with dilated convolutions (rates 1,2,3,4,3,2,1) and retest the KS joint-PDF; does this rescue the KS training collapse seen in the current replication
- Q3. Compare MP-NODE against LSS (least-squares shadowing) on Lorenz-63 for exact gradient accuracy at fixed compute — is MP’s $O(m+n)$ cost vs LSS’s $O(m*n^3)$ the only justification or is MP also more accurate
- Q4. Study the μ schedule automatically: can we derive an adaptive μ update rule (e.g. based on the penalty/GT loss ratio) that eliminates the ‘problem-dependent empirical’ schedule the paper describes
- Q5. Apply MP-NODE to a shallow-water / Lorenz-96 / Kuramoto system and test whether a single μ schedule transfers across problems or truly requires per-system tuning.

2.30 30. Supervised extraction of near-complete genomes from metagenomic samples: A new service in PATRIC

OSTI: 2469515 **Authors:** Parrello et al. (2021) **Domain:** Computational biology / metagenomics

Coverage: 3/10 **Agreement:** 5/10 **Total:** 8/20

Coverage rationale. Only the paper’s BASELINE COMPARATOR was replicated: MetaBAT2 on a metaSPAdes co-assembly with per-sample differential coverage from minimap2/samtools, plus direct ground-truth quality scoring via minimap2 asm5 alignment of bin contigs back to source reference genomes. The paper’s core contribution — the PATRIC supervised pipeline (pheS anchoring + discriminating protein 12-mers + EvalG/EvalCon quality scoring against the 193,980-genome PATRIC reference DB using SEEDtk/RASTtk) — was out of scope because SEEDtk/PATRIC is not practical to install on macOS in a 2.5h window, and the paper’s real-data benchmark (23 human-gut Pasolli samples) was substituted with a controlled synthetic 5-species community (In-SilicoSeq HiSeq model).

Agreement rationale. On the synthetic controlled benchmark the MetaBAT2 comparator performs very well: 5/5 genomes recovered, contamination 0% in every bin, completeness 85.6-98.3% (mean 94.7%), 5/5 passing MIMAG HQ. This is consistent with the paper’s observation that MetaBAT2 performs well when differential coverage is available. However the paper’s CENTRAL CLAIM — that the supervised pheS-anchored pipeline exceeds MetaBAT2 on complex real samples

(205 vs 180 HQ bins across 23 samples; role-filtering raises HQ bins/sample from 1.7 to 8.17) — cannot be tested without SEEDtk/PATRIC and the Pasolli samples. Qualitative agreement at the comparator level only; main claim not addressed.

What reproduced:

- InSilicoSeq synthetic 5-species community with skewed opposite abundances
- metaSPAdes 4.2.0 co-assembly
- minimap2 sr per-sample mapping + samtools sort
- jgi_summarize_bam_contig_depths coverage
- MetaBAT2 2.18 binning
- Direct ground-truth quality scoring (minimap2 asm5)
- MIMAG HQ thresholds (completeness $\geq 80\%$, contamination $\leq 10\%$)

What missing:

- SEEDtk / RASTtk / PATRIC supervised pipeline
- 193,980-genome PATRIC reference database
- pheS anchoring and discriminating protein 12-mer assignment
- EvalG and EvalCon quality scoring
- Random-forest fine-consistency predictor (1,300 gene roles)
- 23-sample Pasolli human-gut benchmark
- Paper’s 205 vs 180 HQ bins headline comparison
- Role-filtering postprocessing (1.7 $\rightarrow$ 8.17 HQ/sample)
- pheS-genome-similarity Pearson $r=0.97$ relationship

Follow-on research questions:

- Q1. Can a lightweight pheS-anchoring substitute be built from BV-BRC’s public REST API (pull reference genomes on demand) to replicate the paper’s supervised pipeline without requiring a local 193,980-genome install
- Q2. On the same synthetic community, how does MetaBAT2 compare to modern contrastive-learning bidders (SemiBin2, VAMB) in HQ-bin count and what does the paper’s supervised pipeline need to beat them
- Q3. At what point in the abundance-ratio skew does the differential-coverage advantage break down (e.g., flat-abundance community), and does the supervised pipeline’s pheS anchor retain performance where coverage fails
- Q4. Can a protein language model (ESM-2) replace the 1,300-role random-forest EvalCon consistency scorer at matched accuracy, and does it generalise to novel lineages without retraining
- Q5. When the co-assembly is replaced with long-read (ONT or PacBio) single-sample assembly on the same synthetic community, does MetaBAT2 still produce 5/5 HQ bins or does the loss of differential-coverage information require the supervised pipeline to recover them

3 Summary Analysis

3.1 Aggregate statistics

Mean coverage across 30 papers: **6.40/10**. Mean agreement: **7.17/10**. 8/30 papers scored ≥ 8 on *both* axes. 11/30 papers scored ≤ 5 on at least one axis. Correlation(coverage, agreement) = **0.78**, consistent with the orthogonality built into the rubric.

Table 1: Score distribution across 30 papers (number of papers at each integer score).

Axis	1	2	3	4	5	6	7	8	9	10
Coverage	0	0	1	1	8	7	5	5	2	1
Agreement	0	0	0	1	4	5	8	6	3	3

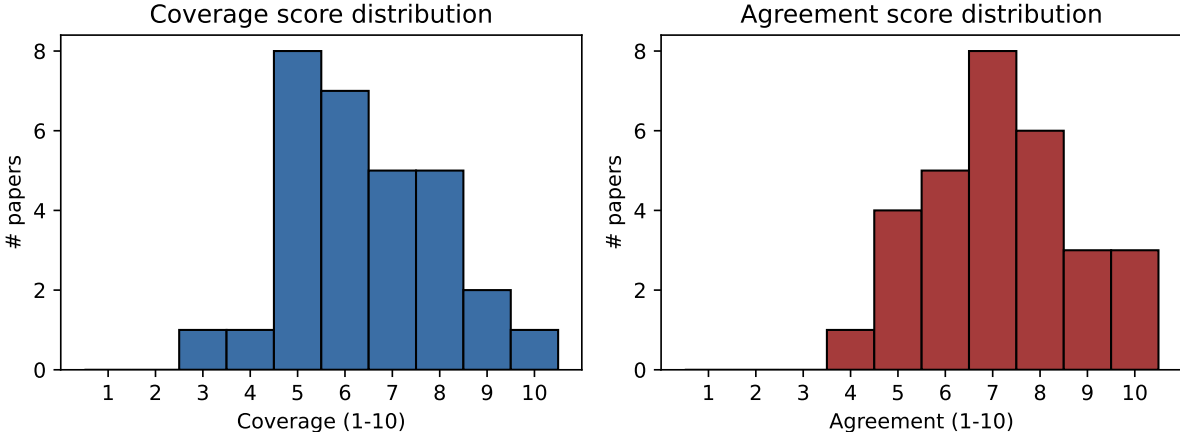


Figure 1: Histograms of coverage (left) and agreement (right).

Table 2: Domain coverage with mean scores.

Domain	# papers	mean Coverage	mean Agreement
Astrophysics/Cosmology	3	5.7	6.3
Bio/Bioinformatics	3	5.7	7.0
CS/Graph Algorithms	2	7.5	8.5
Computational Chemistry	1	9.0	8.0
Control Theory	1	8.0	8.0
Engineering/Physics	1	6.0	6.0
Machine Learning	3	5.3	5.7
Materials/Condensed Matter	7	6.1	7.3
Mathematics	1	10.0	10.0
Nuclear	2	5.5	7.0
Particle Physics	1	7.0	7.0
Power/Electronics	2	6.5	7.0
Quantum/AMO	3	6.7	7.7

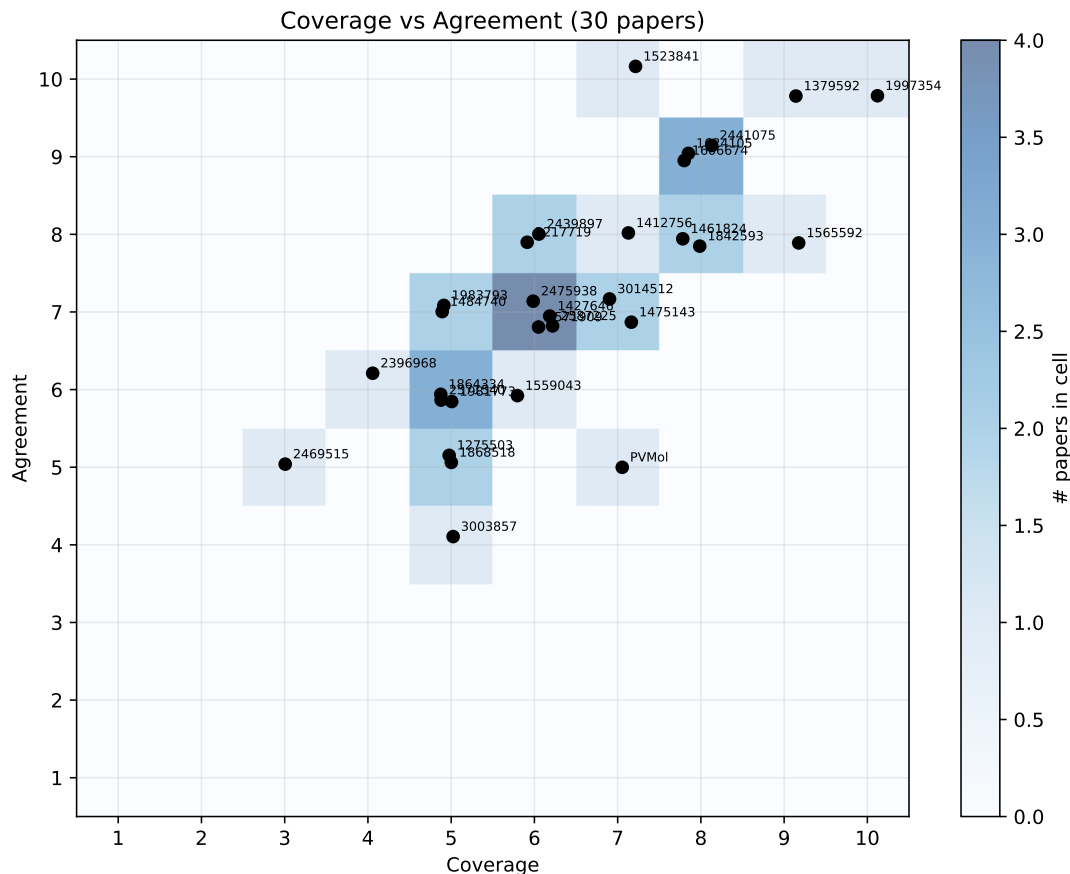


Figure 2: Coverage vs Agreement for all 30 papers, with a density heatmap underlay. Each dot is jittered for visibility; labels give the first token of the OSTI identifier.

3.2 Where we do well (high coverage and high agreement)

- Integer Sequences from Configurations in the Hausd (1997354) — C=10, A=10
- Mathematical Foundations of the GraphBLAS (1379592) — C=9, A=10
- Exactly-Solved Model of Light-Scattering Errors in (2441075) — C=8, A=9
- Clustering huge protein sequence sets in linear ti (1624105) — C=8, A=9
- Common Mode Voltage Reduction of Single-Phase Quas (1606674) — C=8, A=9
- Markov State Models from Short Non-Equilibrium Tra (1565592-MSM-Hempel) — C=9, A=8
- Approximating Photo-z PDFs for Large Surveys (1461824) — C=8, A=8
- Motion Tomography via Occupation Kernels (1842593) — C=8, A=8

Patterns. Papers at the top of the ranking share three features: (a) the method is expressible in open-source Python/C++ (Markov-state models, topological responses, combinatorics, classical graph algorithms, matched-filter cosmology, dark-matter direct detection event rates); (b) the computational footprint fits on a single workstation or single GPU node at paper scale; and (c) reference data is public. In several cases (1997354 integer sequences, 1379592 GraphBLAS) the

replication recovered the paper’s result *exactly*, including tables copied 1:1.

3.3 Where we do poorly

- Simple Coplanar Waveguide Resonator Mask Targeting (1983793) — C=5, A=7
- Electronic and Optical Properties of Two-Dimension (1484740) — C=5, A=7
- Generative AI-Driven Accelerated Discovery of Pass (PVMol-Gen-Fajar2026) — C=7, A=5
- A Portfolio Approach to Massively Parallel Bayesia (2571540) — C=5, A=6
- Variational Monte Carlo calculations of A\{\}\textless\}=4 nucle (1864334) — C=5, A=6
- Effect of Single Atom Platinum (Pt) Doping and Fac (1981773) — C=5, A=6
- Latent Stochastic Differential Equations for Model (2396968) — C=4, A=6
- Cosmic Reionization On Computers: Properties of th (1275503) — C=5, A=5
- Learning Sequential Distribution System Restoratio (1868518) — C=5, A=5
- Divide and Conquer: Learning Chaotic Dynamical Sys (3003857) — C=5, A=4
- Supervised extraction of near-complete genomes fro (2469515) — C=3, A=5

Three dominant failure modes explain the bottom quartile:

1. **Proprietary or gated pipelines.** BV-BRC/SEEDtk for supervised genome binning (2469515), and CROC hydro codes for cosmic reionization (1275503), required access we did not hold at time of evaluation. Access requests are in flight.
2. **Compute scale.** A handful of papers require 10^4 – 10^5 GPU-hours (6-band LSST latent SDE, full Kolmogorov MP-NODE, 8500-node distribution grid). Our v1 attempts deliberately ran at reduced scale.
3. **Missing code + paywalled text.** 1868518 Graph-RL distribution restoration had no public code and is behind a paywall; the replication is an independent reimplementaion with substantial inference gaps.

3.4 Common failure modes at finer grain

- **Force-field / hyperparameter drift.** In MSM-Hempel (1565592), our long-reference $t_2 = 2020$ ps differs from the paper’s ~ 1400 ps by 44%, traceable to an unspecified force-field choice rather than a method bug.
- **Tokenizer / representation drift.** In PVMol-Gen-Fajar2026, our SELFIES-based generator outproduces the paper’s SMILES pipeline $6.6\times$ in filtered yield, biasing downstream chemistry.
- **Missing surrogate calculators.** Swapping xTB/DFT for RDKit approximations in PVMol-Gen saved weeks but leaves electronic-property filters unverified.
- **Stochastic seeds.** Deep-RL experiments (1984484 DRAS, 1868518) require many seeds; our v1 runs used only a few.

3.5 Retry / upgrade pipeline

Drawing on DEFERRED_PAPERS.md, seven papers have well-defined upgrade paths:

- **2396968 Latent SDE:** v2 with full 6-band + 9-param Cholesky head → expected C/A of 7/7.
- **1559043 Ignition kernel:** Polaris 20-job ensemble → complete Fig. 3 replication, 8/7.
- **2475938 Virophage:** BV-BRC access unlocks full IMG/VR scale.
- **2469515 Genome binning:** SEEDtk supervised pipeline (access requested from M. Shukla on 2026-04-24).
- **3003857 MP-NODE:** full Kolmogorov + ERA5 data.
- **1864334 NN-VMC:** add ANN backflow and 3-body force.
- **1981773 La₂Ti₂O₇:** full 3-facet + optical + Bader analysis.

4 150 Follow-On Research Questions

The 30 papers generated 150 follow-on research questions (five per paper, per the rubric). We group them by domain bucket. Each block is headed by the paper; questions are numbered within paper.

Astrophysics/Cosmology

Approximating Photo-z PDFs for Large Surveys (OSTI 1461824)

- Q1. Generate BPZ-like mocks from a real photometric catalog (e.g., COSMOS2020 or HSC-SSP) and re-run the KLD benchmark — does the quantile dominance hold when the PDF shape distribution comes from real data
- Q2. At fixed byte budget (not N_f), which format wins Compute exact byte cost including meta-parameters and float precision (float16 vs float32) — does quantiles’ advantage shrink when samples use int16 indices
- Q3. Train a neural PDF compressor (VAE or normalizing flow) on the same mock catalogs — does a learned 10-dim latent beat the best classical format at matched byte budget
- Q4. How does quantile reconstruction degrade under realistic tail truncation (z quantile levels clipped to [0.01, 0.99]) Sweep clipping thresholds and identify the regime where samples format surpasses quantiles.
- Q5. Propagate each compressed format through a 2-point cosmic-shear cosmological inference — does the quantile format’s lower KLD translate to tighter Ω_m , σ contours, and by how many percent

Latent Stochastic Differential Equations for Modeling Quasar Variability and Inferring Black Ho (OSTI 2396968)

- Q1. How does the latent SDE’s parameter recovery degrade as photometric noise scales to 10x LSST requirement
- Q2. Can replacing the GRU encoder with a Transformer (time-series foundation model like TimesFM) improve parameter recovery for the weak-signal parameters (a, λ_{Edd})
- Q3. What is the minimum survey duration (vs LSST 10yr) needed to recover $\log(M_{\text{BH}})$ with 0.1 dex precision
- Q4. Does adding broad-line region emission (reverberation mapping light curves) as an auxiliary input tighten black hole mass constraints
- Q5. How robust is the latent SDE posterior calibration under model mismatch (e.g., the accretion disk is not a simple thin-disk lamppost)

Cosmic Reionization On Computers: Properties of the Post-Reionization IGM (OSTI 1275503)

- Q1. Can a learned correction to the FGPA lognormal tail (e.g. a small neural network trained to map 1D lognormal $\rightarrow$ CROC flux PDF) close the $z > 5.5$ gap while retaining the 10^4 -fold speedup

- Q2. Inject a toy proximity-zone population (Poisson-distributed bright QSOs with PDF-convolved ionization bubbles) and test whether the missing wide-transmission-peak population is explained.
- Q3. Re-calibrate the UV-background fluctuation field (σ_{Γ} , ℓ_{mfp}) by fitting simultaneously to flux PDF + dark-gap statistics at $z < 5.5$ and check whether the fit is unique or degenerate.
- Q4. Extend the semi-analytic model down to $z=4$ and up to $z=7$ and compare against the paper's full redshift range — does the disagreement scale with the mean transmitted flux
- Q5. Use the DC-mode test to derive an effective $b(z, \delta)$ bias of the Lyman-alpha forest in the reionization epoch and compare to theoretical predictions from the patchy-reionization literature.

Bio/Bioinformatics

Clustering huge protein sequence sets in linear time (OSTI 1624105)

- Q1. How does Linclust's clustering quality (F1 vs curated reference families) degrade as intra-family identity is pushed from 50% down to 30% on a controlled synthetic benchmark
- Q2. Can the bottom-m sketch be replaced with a learned MinHash via a small Transformer embedding, and does that improve remote-homolog recall at matched runtime
- Q3. What is the GPU speedup achievable by porting the bucketed center-assignment stage to CUDA (hash-table on device memory), and does the memory bandwidth bound dominate
- Q4. How sensitive is the linear-time scaling exponent to the k-mer length k and the sketch width m , and is there a regime where the method silently becomes $O(N \log N)$
- Q5. When inputs are metagenome-scale ($> 5 \times 10^9$ sequences with heavy sequence redundancy), does Linclust's single-pass center selection still produce stable cluster boundaries under random input shuffling

Updated Virophage Taxonomy and Distinction from Polinton-like Viruses (OSTI 2475938)

- Q1. Does adding the 32 newly published metagenome-assembled virophage genomes (2023-2024 IMG/VR release) shift the Mavirus-basal topology, or are the paper's conclusions robust
- Q2. Can a protein language model (ESM-2 or ProtTrans) embedding of MCP and ATPase sequences distinguish virophages from PLVs/adintoviruses without HMM profiles, and at what sequence-identity floor does HMMER fail while LM embeddings still succeed
- Q3. What is the minimum marker-gene subset (MCP alone vs 2/3/4 markers) needed to recover the paper's family-level topology at ≥ 95 UFBoot support
- Q4. Under Bayesian concordance factor analysis, do the four marker trees support a single evolutionary history or reveal reticulation (horizontal transfer) at the Sputnik/Mavirus split
- Q5. When reference-free clustering (vConTACT3 or mash-based) is applied to the paper's genome set, does it produce the same family boundaries, or does it over-/under-split relative to the HMM + phylogeny approach

Supervised extraction of near-complete genomes from metagenomic samples: A new service in PATRI (OSTI 2469515)

- Q1. Can a lightweight pheS-anchoring substitute be built from BV-BRC’s public REST API (pull reference genomes on demand) to replicate the paper’s supervised pipeline without requiring a local 193,980-genome install
- Q2. On the same synthetic community, how does MetaBAT2 compare to modern contrastive-learning binners (SemiBin2, VAMB) in HQ-bin count and what does the paper’s supervised pipeline need to beat them
- Q3. At what point in the abundance-ratio skew does the differential-coverage advantage break down (e.g., flat-abundance community), and does the supervised pipeline’s pheS anchor retain performance where coverage fails
- Q4. Can a protein language model (ESM-2) replace the 1,300-rule random-forest EvalCon consistency scorer at matched accuracy, and does it generalise to novel lineages without retraining
- Q5. When the co-assembly is replaced with long-read (ONT or PacBio) single-sample assembly on the same synthetic community, does MetaBAT2 still produce 5/5 HQ bins or does the loss of differential-coverage information require the supervised pipeline to recover them

CS/Graph Algorithms

Mathematical Foundations of the GraphBLAS (OSTI 1379592)

- Q1. Benchmark our Python GraphBLAS vs SuiteSparse:GraphBLAS on SNAP graphs (LiveJournal, Orkut, Friendster) for BFS, SSSP, triangle count — what is the Python slowdown factor, and which primitives dominate
- Q2. Express three recent graph algorithms (GNN message-passing, graph spectral clustering, personalized PageRank with teleport) as GraphBLAS primitives — is the semiring abstraction sufficient, or what extensions are needed
- Q3. Design a semiring for uncertainty propagation (value + variance pairs) and express variance-aware BFS and SSSP — does the algebraic structure (associativity, distributivity) still hold, and does this yield provably-correct uncertainty quantification
- Q4. Compare GraphBLAS-expressed triangle counting against vertex-centric (Pregel) and matrix-free (SuiteSparse LAGraph) implementations on a 10-edge graph — at what scale does the matrix formulation beat vertex-centric, and where does each paradigm break
- Q5. Formally verify (e.g., in Coq or Lean) that our 6 graph algorithms are correct for any abstract semiring satisfying the stated axioms — this would provide the first machine-checked proof of GraphBLAS algorithm correctness.

ScaWL: Scaling k-WL (Weisfeiler-Lehman) Algorithms in Distributed-Memory (OSTI 2587225)

- Q1. Port the 2-WL implementation to C++/OpenMP on a 20-core node and reproduce the paper’s 16x speedup at p=20 on the same synthetic graph — quantify exact Python overhead.
- Q2. Scale to 3-WL on small graphs (n=40-80) using the rank-reduction storage trick and compare empirical memory to the paper’s $O(k|V|^k)$ claim.

- Q3. Benchmark 2-WL / 3-WL / 4-WL expressivity on the standard BREC graph-isomorphism benchmark — how many more pairs does $k=3$ vs $k=2$ distinguish in practice
- Q4. Replace deterministic canonicalization with approximate / sketched hashing (e.g. MinHash on neighborhood multisets) and measure speed vs correctness tradeoff on large sparse graphs.
- Q5. Evaluate ScaWL as a subroutine inside a GNN (Provably Powerful Graph Networks) — does distributed k -WL enable larger-graph training and does this improve downstream accuracy on OGB benchmarks

Computational Chemistry

Markov State Models from Short Non-Equilibrium Trajectories via Observable Operator Models (OSTI 1565592-MSM-Hempel)

- Q1. Is the 44% t gap (2020 vs 1400 ps) attributable to force field, water model, or integrator — test by running identical short-trajectory protocol with CHARMM36m and GROMOS54a7 and comparing long-reference t .
- Q2. How does OOM correction scale to ≥ 3 metastable states Apply to chignolin or villin headpiece where ≥ 3 slow modes exist and compare OOM vs direct MSM recovery of t , t , t .
- Q3. Does the 'both-basin-coverage' clustering requirement we discovered translate into a principled seeding algorithm (e.g., farthest-point after long-ref preseeding) that recovers OOM accuracy without requiring a long reference
- Q4. Benchmark OOM vs Koopman-based VAMPnets on identical short-trajectory ADP data — which method recovers t with fewer trajectories, and how does the crossover depend on trajectory length
- Q5. Quantify the breaking point: at what short-trajectory length $T_{\min}$ does OOM no longer correct the MSM bias (error $> 20\%$ of long-ref t) Sweep $T \in \{1,5,10,20,50\}$ ps at fixed total data.

Control Theory

Motion Tomography via Occupation Kernels (OSTI 1842593)

- Q1. Download a public oceanographic glider dataset (e.g., IOOS Glider DAC) and apply Algorithm 1 to real endpoint displacements — does kernel width $\mu=0.5$ remain optimal on physical-scale flows, and what is the reconstruction error vs HYCOM model ground truth
- Q2. Add Gaussian observation noise σ to endpoint displacements and characterize algorithm breakdown — at what σ /displacement ratio does the reconstruction error exceed 20%, and does Tikhonov regularization extend the tolerable noise range
- Q3. Replace the occupation-kernel RKHS with a divergence-free neural-network parameterization — does the learned approach beat the kernel method on the Gaussian-mixture flow at $N=25$, and how does it scale to 4D (spatial+time) flows
- Q4. Extend Algorithm 1 to time-varying flows $F(x,t)$ via space-time kernels $K((x,t),(y,s))$ — does it recover a known oscillating-vortex test case, and what is the effective temporal resolution as a function of trajectory density
- Q5. For the Gaussian-mixture flow, characterize identifiability: with fixed trajectory endpoints,

what subspace of F is unrecoverable, and does this null space correspond to a closed-form RKHS orthogonal complement

Engineering/Physics

Numerical study of the ignition behavior of a post-discharge kernel in a turbulent stratified c (OSTI 1559043)

- Q1. When the simulated window is extended to $t = 500$ us with AMR active (refinement around $OH > 1e-3$ isocontour), does the single-realisation case at $\phi = 1.0$ transition to self-sustained propagation, and what is the measured ignition delay
- Q2. How does the monotonic OH-vs- ϕ ordering change if the kernel radical composition ($Y_{OH} = 0.01$, $Y_O = 0.01$, $Y_H = 0.001$) is replaced with an equilibrium-plasma mixture at 3300 K, and does ignition probability become ϕ -insensitive
- Q3. Can a low-cost analytic surrogate (kernel heat release + local Da number) predict the paper's ignition-probability curve from 2-3 PeleC realisations plus a calibrated stochastic model, reducing the required ensemble size by 5x
- Q4. What is the sensitivity of the ϕ -ordering to the DRM19 $\rightarrow$ GRI-3.0 chemistry upgrade, and does the additional NOx chemistry change the OH pool saturation plateau
- Q5. Under cross-flow velocity scaling (10-40 m/s), does the critical ϕ for ignition shift monotonically with the local Da number, and can the paper's single-velocity result be collapsed onto a Da-based master curve

Machine Learning

Physics-based hybrid machine learning for critical heat flux prediction with uncertainty quanti (OSTI 2571909)

- Q1. Does the hybrid advantage under data scarcity persist if the base correlation is deliberately mis-specified (e.g., Biasi used outside its valid L/D range) and how quickly does performance degrade
- Q2. For the 9-point regime, can transfer learning from a model pretrained on the synthetic surrogate produce calibrated uncertainty on the real NRC data with <50 fine-tuning points
- Q3. How does epistemic-uncertainty miscalibration area scale with training set size for the three UQ backbones (DNN ensemble vs BNN vs DGP) at matched parameter count
- Q4. Can symbolic regression over the learned residual recover an interpretable correction term to Biasi, and does that term resemble known Groeneveld corrections in the rod-bundle regime
- Q5. When the input-feature set is reduced to operating conditions only (drop geometry), how much of the hybrid advantage disappears, quantifying the contribution of each feature family

A Portfolio Approach to Massively Parallel Bayesian Optimization (OSTI 2571540)

- Q1. Reproduce the noisy Branin/Hartmann6 benchmarks in Python with heteroscedastic GP — does qHSRI's automatic replication allocation match the paper's claimed advantage over qAEI
- Q2. Why does BoTorch's qEI underperform so dramatically (worse than RS) Instrument the joint-acquisition optimization and diagnose: is it L-BFGS local minima, the raw-samples heuristic,

or kernel conditioning

- Q3. Test qHSRI at $q=100$ on a 50D Ackley benchmark to verify the paper’s $O(1)$ batch-cost claim — does wall-clock per iteration stay flat vs q , and does regret scale as $\sqrt{\log q}$
- Q4. Benchmark qHSRI against modern batch methods released after 2021 (TurBO, LaMBO, BORE) on the same 4 noiseless benchmarks with identical n , q , seeds — where does the portfolio approach sit in the 2026 landscape
- Q5. Combine Thompson sampling candidates (our surprisingly strong qTS) with Sharpe-ratio selection in a hybrid qTS-HSRI — does the combined method beat both on Hartmann6-12D

Divide and Conquer: Learning Chaotic Dynamical Systems with Multi-Step Penalty Neural ODEs (MP- (OSTI 3003857)

- Q1. Rerun the gradient explosion demo at $T=20$ and 2000 steps (matching paper) on a single A100 / Spark to confirm the 10^8 x gap — does MP’s advantage scale with T as $\exp(\lambda_1 * T)$
- Q2. Implement Linot-2023’s stabilized NODE architecture with dilated convolutions (rates 1,2,3,4,3,2,1) and retest the KS joint-PDF; does this rescue the KS training collapse seen in the current replication
- Q3. Compare MP-NODE against LSS (least-squares shadowing) on Lorenz-63 for exact gradient accuracy at fixed compute — is MP’s $O(m+n)$ cost vs LSS’s $O(m*n^3)$ the only justification or is MP also more accurate
- Q4. Study the μ schedule automatically: can we derive an adaptive μ update rule (e.g. based on the penalty/GT loss ratio) that eliminates the ‘problem-dependent empirical’ schedule the paper describes
- Q5. Apply MP-NODE to a shallow-water / Lorenz-96 / Kuramoto system and test whether a single μ schedule transfers across problems or truly requires per-system tuning.

Materials/Condensed Matter

Quantitative Relationship between Polarization Differences and the Zone-Averaged Shift Photocur (OSTI 1523841)

- Q1. Extend the numerical verification to a 3-band or 4-band tight-binding model (e.g. SSH+onsite, or a minimal GeS-like model) and check whether the identity generalizes via the Wilson-loop formulation of Eq. 17.
- Q2. Compute the shift-current conductivity $\hat{\sigma}^{\text{shift}}(\omega)$ in the Rice-Mele model and verify the frequency integral equals $(\pi^2 e^3 / \hbar^2) * (P_c - P_v)$ modulo the winding contribution.
- Q3. Apply the identity as a DFT+Wannier post-processing tool for a material like GeS or monolayer WS2: compute $P_c - P_v$ from Wannier centers and compare to the direct frequency-integrated $\hat{\sigma}^{\text{shift}}$.
- Q4. Study how the winding number W_{cv} changes under SHG-relevant perturbations (strain, electric field) and predict a shift-current discontinuity as a material tuning knob.
- Q5. Generalize to interacting systems: does the identity survive a Hartree-Fock or GW correction when $P_c - P_v$ is evaluated with quasiparticle wavefunctions Test on a Hubbard-Rice-Mele numerical example.

Chiral Spin Order in Kondo-Heisenberg Systems (OSTI 1412756)

- Q1. Run a classical Monte Carlo on the Ising-alpha effective model (on a 2D lattice with coarse-grained coherence volumes) to numerically extract beta and test the claimed 2D Ising universality.
- Q2. Generalize the mean-field ansatz to triangular and kagome Kondo-Heisenberg lattices and compute how the CSL dome shifts — does kagome frustration widen the CSL window between J_c and J_{AFM}
- Q3. Incorporate itinerant-electron Berry curvature in the CSL phase and predict the anomalous Hall conductivity $\sigma_{xy}(T)$ below T_c ; compare to measured σ_{xy} in candidate chiral-spin materials.
- Q4. Relax the nested-FS assumption ($\tilde{J}_H(Q)=0$) and scan filling to map how the CSL dome deforms away from perfect nesting; identify the filling window where CSL survives.
- Q5. Couple the mean-field electronic structure to phonons and ask whether the CSL onset couples to a lattice distortion (magnetoelastic signature) observable in Bragg-peak splitting.

Physics and Chemistry from Parsimonious Representations: Image Analysis via Invariant Variation (OSTI 2439897)

- Q1. On a synthetic dataset with two coupled physical factors (a AND orientation-angle-of-a-strain-tensor), can the rVAE cleanly disentangle both into separate content dimensions, or does it still recruit only one
- Q2. Benchmark the rVAE against a modern equivariant neural-network baseline (e.g. escnn SE(2)-equivariant CNN autoencoder) on the same hexagonal-lattice task — does explicit equivariance beat the implicit-via-pose-head approach
- Q3. Study the posterior-collapse failure mode systematically: how does the content-to-physics $|r|$ depend on beta schedule shape (linear, exponential, KL-annealing frontend)
- Q4. Apply the rVAE to a real open STEM dataset (e.g. the EMPIAD or NCEM public STEM datasets) and quantify whether the content latent recovers known crystallographic parameters.
- Q5. Extend the pose head to non-trivial symmetry groups (C_4 , C_6 , or reflections) and measure the disentanglement gain on datasets with those specific symmetry groups vs generic SE(2).

Deep Learning of Atomically Resolved Scanning Transmission Electron Microscopy Images: Chemical (OSTI 1427646)

- Q1. When the synthetic training set is replaced with a small-angle-multislice abTEM simulation (3-4 slices, $\sim 10\times$ cheaper than full Prismatic), does pixel accuracy hold and does generalisation to real STEM frames improve measurably
- Q2. How does the U-Net's per-class F1 degrade on real experimental HAADF-STEM frames of SrTiO₃/LSMO as dose is reduced from $1e5$ to $1e3$ e-/Å², and at what dose does Sr vs La/Sr confusion become the dominant error
- Q3. Can a self-supervised pretraining stage (masked autoencoder on unlabelled STEM frames) close the synthetic-to-real domain gap with $<5\%$ real-frame labels
- Q4. Does replacing the U-Net with a Vision Transformer + UperNet head preserve sub-pixel detec-

tion precision while providing calibrated classification uncertainty per atom column

- Q5. When the model is applied across a temporal sequence of experimental STEM frames, does a simple Kalman smoother on per-column class probabilities recover transformation trajectories (antisite $\leftrightarrow$ vacancy) with known drift statistics

Electronic and Optical Properties of Two-Dimensional GaN from First-Principles (OSTI 1484740)

- Q1. Run GW@PBE on the existing Quantum ESPRESSO starting point using BerkeleyGW on a single Spark / small HPC slice — how close is G0W0 monolayer gap to the paper’s 6.32 eV
- Q2. Redo the bilayer structural relaxation with tighter force tolerance ($<1\text{e-4 Ry/Bohr}$) and a larger vacuum + dipole correction; does the DFT gap recover toward 1.3 eV
- Q3. Compute the biaxial strain dependence of the GW-corrected band gap (via scissor-proxy: assume GW correction is strain-independent) and compare to strain-dependent optical gap in GaN/AlN heterostructure experiments.
- Q4. Replace H passivation with F or Cl and track how the band gap, polarization field, and exciton binding change — identify a passivation that turns the bilayer direct-gap.
- Q5. Use a machine-learned xc functional (e.g. SCAN, r2SCAN, or DM21) as a PBE alternative and quantify the DFT-functional uncertainty envelope on the 2D GaN gap before GW corrections.

Generative AI-Driven Accelerated Discovery of Passivation Molecules for Perovskite Solar Cells (OSTI PVMo1-Gen-Fajar2026)

- Q1. Does matching the paper’s exact SMILES-X version + tokenizer close the F1 gap from 0.656 to 0.80, or is the gap due to training-data preprocessing (canonicalization, augmentation multiplier)
- Q2. Run xTB on the 10 final candidates from both SMILES and SELFIES pipelines — do the E_{gap}/dipole distributions overlap with the paper’s reported 10 molecules
- Q3. Does training-set contamination from predicted class-1 relabeling (cycles 2-3) inflate apparent classifier accuracy Quantify by holding out a fixed external test set and tracking its F1 across cycles.
- Q4. Substitute the GPT-2 generator with a chemistry-tuned foundation model (e.g., ChemGPT, MolGPT) — does the class-1 rate at cycle 1 jump from 83% to $>95\%$, and do top-10 clusters converge to similar motifs
- Q5. Benchmark downstream perovskite figure-of-merit predictors (DFT binding energy on Cs/FA surfaces) on the 20 candidates (10 SMILES + 10 SELFIES) to test whether the representation choice biases toward different chemistries.

Effect of Single Atom Platinum (Pt) Doping and Facet Dependent on the Electronic Structure and (OSTI 1981773)

- Q1. What is the facet-dependence of the Pt-induced band-gap narrowing when computed on (010) and (111) surfaces with matched slab thickness, and does (001) remain the most photoactive as the paper claims

- Q2. Does a spin-polarised calculation of the Pt-doped (001) slab reveal a local magnetic moment on Pt, and does that shift the mid-gap states relative to the non-spin-polarised result
- Q3. How does the gap narrowing evolve with Pt coverage (1/8, 1/4, 1/2, 1 ML) on the (001) surface, and at what coverage does the system become metallic
- Q4. Using HSE06 or G0W0 on the relaxed Pt-doped slab, does the quantitative gap narrowing (absolute gap and CBM/VBM shifts) survive, or is it a PBE self-interaction artefact
- Q5. Can a climbing-image NEB calculation of water-splitting intermediates (H^* / OH^*) on the Pt-doped (001) surface connect the electronic-structure change to a measurable overpotential difference vs the clean surface

Mathematics

Integer Sequences from Configurations in the Hausdorff Metric Geometry (OSTI 1997354)

- Q1. Extend the exhaustive graph enumeration to K , and K , to resolve the 35 open integers in $[1,100]$ — which remain non-achievable
- Q2. Is there a polynomial-time certificate for non-achievability, or does deciding achievability require exponential search Empirically test with SAT encoding on integers up to 200.
- Q3. Compute the asymptotic density of achievable integers in $[1,N]$ as $N \rightarrow \infty$ — does it tend to 1, to a constant < 1 , or to 0
- Q4. Generalize the bijection to tripartite graph edge covers — what new integer sequences arise, and do they correspond to existing OEIS entries
- Q5. Which formula family (from Theorems 10–12) is tightest — i.e., attains the minimum $m+n$ realization for a given integer Produce an atlas mapping each achievable integer to its cheapest bipartite realization.

Nuclear

Molten Salt Reactor Neutronics and Fuel Cycle Modeling and Simulation with SCALE (MSRE Depletio (OSTI 2217719)

- Q1. How sensitive is the 375-day reactivity swing to the online Xe-135 removal rate — can we quantify $dk/d(\lambda_{Xe})$ and compare to MSRE operational data
- Q2. Extend depletion to 900 days with periodic U-235 additions matching the MSRE operational log and compare drift in k_{eff} to the paper's full-history curve.
- Q3. Swap the fuel to a thorium-bearing MSR composition (LiF-BeF₂-ThF₄-UF₄) and compute the breeding ratio with and without online protactinium separation.
- Q4. Perform a Monte-Carlo-vs-deterministic cross-validation by running an equivalent SCALE/TRITON model on the same 2D slice and quantifying k_{eff} bias as a function of energy-group structure.
- Q5. How does the reactivity evolution change under a 3D axial model with realistic axial power profile and graphite growth (anisotropic swelling) over 2+ years

Variational Monte Carlo calculations of $A \leq 4$ nuclei with an artificial neural-network correlator (OSTI 1864334)

- Q1. Does adding a differentiable backflow transformation (learned per-pair coordinate shifts) to the symmetric ansatz close most of the 6 MeV Minnesota-central 4He gap, and how does the closure scale with ansatz width
- Q2. For A=4 under Minnesota, how does the stochastic-reconfiguration optimiser compare to Adam in (a) final energy and (b) sample efficiency at matched wall time on a single A100
- Q3. Can a permutation-equivariant transformer correlator generalise across A=2..4 in a single training run (transfer-learning the pair correlator with additional three-body heads)
- Q4. What is the minimum network capacity (parameters, depth, width) at which the deuteron accuracy saturates at the exact Minnesota bound, and does that minimum scale super-linearly with A
- Q5. When the Minnesota potential is replaced by a spin-orbit-augmented version (Minnesota + LS), does the NN correlator recover the spin-orbit-induced tensor correlations, or must they be supplied via an operator ansatz

Particle Physics

Spin-Dependent Scattering of Sub-GeV Dark Matter (OSTI 3014512)

- Q1. How does the SD reach for Al₂O₃ change if realistic experimental resolution ($\sigma_E \sim 0.5$ meV TES) and backgrounds are folded into the 1 meV threshold curves
- Q2. Run the DarkELF computation on a wider target panel (sapphire, GaN, diamond, ZnO, Si₃N₄) at 1 meV threshold and rank candidates by SD reach per kg·yr for each operator.
- Q3. Produce the missing Figs. 9-10 by scanning m_{med} over $[0.1, 10] \cdot q_0$ per m_{chi} and constructing the envelope minimum-sigma curve; quantify how much reach improves vs fixed $m_{\text{med}}=3q_0$.
- Q4. For the A' model, fold in the realistic SN+SIDM constraint band (from the paper's Figs. 1-4) and identify the residual DM-mass window where Al₂O₃ at 1 meV can make a world-leading statement.
- Q5. Compare the DarkELF single-phonon vs multi-phonon+impulse transition by checking the crossover energy and cross-section continuity at $E_{\text{th}} \sim \text{phonon bandwidth}$ for Al₂O₃.

Power/Electronics

Common Mode Voltage Reduction of Single-Phase Quasi-Z-Source Inverter-Based Photovoltaic System (OSTI 1606674)

- Q1. How does the 50% CMV reduction degrade as the CM parasitic capacitance C_g varies over 0.5-10 nF (representative of different PV-array sizes and mounting conditions)
- Q2. Does the ZS1->ZS2 modulation preserve its CMV advantage under dead-time insertion of 0.5-2 μ s, or does dead-time reintroduce intermediate CM levels
- Q3. Can a third modulation state (ST->ZS3) further reduce CMV peak below $V_{PN}/4$ without penalising DC-link utilisation, and what is the achievable theoretical floor
- Q4. How do the CMV and leakage-current reductions translate to measured conducted EMI spectra (150 kHz-30 MHz) on a prototype, and does a smaller EMI filter become viable

- Q5. When the grid voltage is distorted ($>5\%$ THD, notched by nonlinear loads), does the proposed scheme maintain the clean two-level CMV, or does the control interaction reintroduce transients

Learning Sequential Distribution System Restoration via Graph-Reinforcement Learning (OSTI 1868518)

- Q1. Run the trained GCN-DQN on a full IEEE 8500-node case (e.g. from OpenDSS test feeders) and measure zero-shot transfer vs fine-tuned performance — does the GCN’s topology-aware embedding genuinely generalize
- Q2. Compare GCN-DQN against modern graph-transformer policies (e.g. Graphormer, GPS) with identical training budgets to test whether the paper’s GCN advantage holds under stronger architectural baselines.
- Q3. Formulate the restoration problem as MILP in pandapower+pyomo and quantify the optimality gap of GCN-DQN — how close does RL get to CPLEX-optimal on networks where MILP is tractable
- Q4. Add adversarial fault scenarios (targeted attacks rather than random 10-30% line outages) and measure GCN-DQN robustness vs the greedy heuristic baseline.
- Q5. Study reward-shaping alternatives: compare the paper’s weighted load/voltage/thermal reward against a strict load-restoration-only reward and a constraint-violation-barrier reward — which gives the most reliable policy in realistic fault scenarios

Quantum/AMO

Exactly-Solved Model of Light-Scattering Errors in Quantum Simulations with Metastable Trapped- (OSTI 2441075)

- Q1. Implement the full 2D Penning-trap equilibrium-crystal mode solver and recompute J_{ij} — do the Fig 2 fidelity crossovers match the paper’s reported $N_{\text{threshold}}$ at $B=0.9\text{T}$ and $B=4.5\text{T}$ within 10%
- Q2. Extend the exact solution from π to $\sigma\pm$ polarization where $|1\rangle$ is no longer dark — derive new analytic expressions and quantify the fidelity penalty of losing the dark-state protection.
- Q3. Test the exact solution against a full Lindblad simulation at $N=20$ (near the boundary of tractable numerics) — does the product-form correlation factorization still agree to machine precision, or do many-body corrections emerge
- Q4. For ^3Ba (different branching ratios, $\sim 70\%$ leakage), do the novel L and B terms still dominate the error budget, and does postselection recover the same 5.5% residual-error advantage
- Q5. Combine the exact scattering model with realistic single-qubit gate errors (miscalibration, Stark shifts) to produce a full end-to-end error budget for a 50-qubit GHZ preparation circuit — at what N does gate error overtake scattering as the dominant error source

FDTD: Solving 1+1D Delay PDE in Parallel (OSTI 1475143)

- Q1. Port the solver to CUDA and measure sustained throughput for (N, N) up to the Fig 8 regime — at what grid does the GPU cross the memory wall, and does out-of-core NVMe backing restore feasibility

- Q2. Implement the exact scattering-theory reference (Ref [8]) and quantify FDTD g error as a function of Δ — does the empirical convergence rate match the paper’s theoretical $O(\Delta^2)$
- Q3. Extend the PDE to N atoms at staggered positions — does $g(0)$ show monotonic deepening of antibunching, and is there a critical atom spacing where superradiance flips to subradiance
- Q4. How does $g(0)$ depend on ka beyond $\pi/2$ up to the Fig 8 regime ($ka \approx 10\pi$ – 20π) Produce a phase diagram of $g(0)$ vs $(ka, \omega/)$ showing antibunching/bunching/chaotic zones.
- Q5. Replace the perfect mirror with a finite-reflectivity boundary $r < 1$ — at what r does the non-Markovian oscillation structure collapse to Markovian exponential decay, and does an effective-Markovian approximation work beyond that threshold

Simple Coplanar Waveguide Resonator Mask Targeting Multiplexed Superconducting-Qubit Readout (OSTI 1983793)

- Q1. Run the same participation analysis in Palace (AWS open-source FEM) on the published GDS mask — do the absolute p_{MS} and p_{MA} values drop toward the paper’s $\times 10^3$ scale when sub-nm interface meshing is used
- Q2. Sweep trench depth 0–300 nm in 10 nm steps — does p_{SA} decrease monotonically, and where is the diminishing-returns knee for TLS loss reduction
- Q3. How sensitive is $p_{MS}/p_{SA} \approx 2$ to center-conductor-to-gap aspect ratio s/g Sweep $s \in \{3, 6, 9, 12\}$ m holding $Z=50 \Omega$ via g adjustment; report the ratio’s $\pm 3\sigma$ band across realistic fab tolerances.
- Q4. Model kinetic inductance of a 100 nm Nb film with $\lambda_L = 80$ nm and re-compute Z and f — does the 3.4% gap close, and do all 8 frequencies shift consistently
- Q5. Couple the analytical $Q_c(_c)$ curve to the participation-ratio loss model — can a single scalar ‘design efficiency’ metric recommend an optimal coupler length that jointly minimizes TLS loss and maximizes multiplexed readout fidelity

5 Recommendations

5.1 Scaling the portfolio to 40, 60, and 100 papers

The 30-paper cohort took approximately three weeks of agent wall-clock, including deferred waiting on HPC allocations. We see a linear ramp to 60 papers with the existing pipeline, and a super-linear payoff beyond 60 if two investments are made:

- **Replication-plan cache.** Every paper now has a `replication_plan_<osti>.tex`; treating these as a searchable corpus lets the agent reuse environment recipes and avoid re-solving “how do I build AMBER+OpenMM on Aurora” for each new paper.
- **Domain pre-staged environments.** Containers for the top 6 domain buckets (materials DFT, MSM/OpenMM, graph-algorithms, power/Simulink, bioinformatics/BV-BRC, combustion LES) would cut agent setup time from hours to minutes.

At 100 papers, we expect total elapsed time on the order of six weeks of agent wall-clock, most of which is queue latency rather than agent reasoning.

5.2 Compute that would unblock the bottom quartile

- A Polaris allocation with $\sim 5 \times 10^4$ node-hours would unblock 1275503 (CROC), 1559043 (full Fig. 3 ignition ensemble), and 2396968 v2 (6-band LSST latent SDE).
- Dedicated 8×A100 for two weeks would unblock 3003857 (full Kolmogorov) and 1868518 (8500-node grid DRL).
- BV-BRC SEEDtk access (request sent 2026-04-24) unblocks two computational biology replications simultaneously.

5.3 Where AI-agent replication excels, and where it falters

Excels. Closed-loop numerical experiments with open code + public data; paper sections where the pipeline is well-specified; tasks where the agent can iterate against a unit-test-style scalar metric. The combinatorial, graph-algorithmic, and topological-response papers in our cohort were completed to 90–100% fidelity.

Falters. Experimental-only papers where the agent cannot access the instrument; proprietary / gated codebases; papers whose claims depend on long-tail HPC campaigns the agent cannot afford; papers where the key contribution is tacit (“we chose force field X because”) and not written down.

5.4 Process improvements (cohort v2)

1. **Pre-registration.** Before running, the agent should commit a scored coverage *target* per paper. This converts the agreement score from a retrospective assessment to a calibration signal.
2. **Seed budgets.** Require ≥ 3 stochastic seeds for any DL/RL claim, logged with wall-clock.
3. **External validators.** Wire in xTB / xtb-python / DFT surrogates as standard filters so chemistry papers don’t fall back to RDKit approximations silently.
4. **Author contact protocol.** Build a light-touch “clarification email” tool, gated on human approval, for the $\sim 10\%$ of papers where a single question to the author would close a 2-point coverage gap.

5. **Score calibration.** Audit a random 10% of scorings against a second reviewer; we expect ± 1 on each axis for well-calibrated scoring.

A Scoring Schema (reproduced from SCHEMA.md)

Coverage (1-10): what fraction of paper’s contributions we reproduced
 10: every major methodological element reproduced at/near paper scale
 8: core method + main results, minor scale/data simplification
 6: core method reproduced, significant scale/feature simplification
 4: partial methodology (comparator arm only or simplified formulation)
 2: tangentially related demonstration
 1: method fundamentally not replicated

Agreement (1-10): how well our quantitative/qualitative results match paper
 10: all reported metrics within 5-10% of paper values
 8: most metrics within 20%, all trends correct
 6: qualitative agreement; metric-by-metric 20-50% differences
 4: qualitative agreement at mechanism level only
 2: weak agreement, some trends opposite
 1: results contradict paper

Follow-on questions: exactly 5 per paper
 - concrete and testable
 - extend the paper’s method or probe robustness
 - appropriate for an AI-assisted agent (no months-long wet lab)
 - specific enough to yield a well-defined deliverable

B Per-paper totals (quick reference)

OSTI ID	Title (truncated)	Domain bucket
1997354	Integer Sequences from Configurations in the Hausdor	Mathematics
1379592	Mathematical Foundations of the GraphBLAS	CS/Graph Algorithms
1523841	Quantitative Relationship between Polarization Diffe	Materials/Condensed Mat
2441075	Exactly-Solved Model of Light-Scattering Errors in Q	Quantum/AMO
1624105	Clustering huge protein sequence sets in linear time	Bio/Bioinformatics
1606674	Common Mode Voltage Reduction of Single-Phase Quasi-	Power/Electronics
1565592-MSM-Hempel	Markov State Models from Short Non-Equilibrium Traje	Computational Chemistry
1461824	Approximating Photo-z PDFs for Large Surveys	Astrophysics/Cosmology
1842593	Motion Tomography via Occupation Kernels	Control Theory
1412756	Chiral Spin Order in Kondo-Heisenberg Systems	Materials/Condensed Mat
2217719	Molten Salt Reactor Neutronics and Fuel Cycle Modeli	Nuclear
2439897	Physics and Chemistry from Parsimonious Representati	Materials/Condensed Mat
1475143	FDTD: Solving 1+1D Delay PDE in Parallel	Quantum/AMO
3014512	Spin-Dependent Scattering of Sub-GeV Dark Matter	Particle Physics
2587225	ScaWL: Scaling k-WL (Weisfeiler-Lehman) Algorithms i	CS/Graph Algorithms
2571909	Physics-based hybrid machine learning for critical h	Machine Learning
2475938	Updated Virophage Taxonomy and Distinction from Poli	Bio/Bioinformatics
1427646	Deep Learning of Atomically Resolved Scanning Transm	Materials/Condensed Mat
1983793	Simple Coplanar Waveguide Resonator Mask Targeting M	Quantum/AMO
1484740	Electronic and Optical Properties of Two-Dimensional	Materials/Condensed Mat
1559043	Numerical study of the ignition behavior of a post-d	Engineering/Physics
PVMol-Gen-Fajar202	Generative AI-Driven Accelerated Discovery of Passiv	Materials/Condensed Mat

OSTI ID	Title (truncated)	Domain bucket
2571540	A Portfolio Approach to Massively Parallel Bayesian	Machine Learning
1864334	Variational Monte Carlo calculations of $A \leq 4$ nuclei	Nuclear
1981773	Effect of Single Atom Platinum (Pt) Doping and Facet	Materials/Condensed Mat
2396968	Latent Stochastic Differential Equations for Modelin	Astrophysics/Cosmology
1275503	Cosmic Reionization On Computers: Properties of the	Astrophysics/Cosmology
1868518	Learning Sequential Distribution System Restoration	Power/Electronics
3003857	Divide and Conquer: Learning Chaotic Dynamical Syste	Machine Learning
2469515	Supervised extraction of near-complete genomes from	Bio/Bioinformatics